Final Project Report

Woodcraft



Project Supervisor
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Submitted By

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CERTIFICATE

This is to certify that Noor Ul Nisa (B**********) have worked on and completed their Software Project at Software & Research Projects Section, Department of Computer Sciences, Virtual University of Pakistan in partial fulfillment of the requirement for the degree of BS in Computer Sciences under my guidance and supervision.

In our opinion, it is satisfactory and up to the mark and therefore fulfills the requirements of BS in Computer Sciences.

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(Signature)

External Examiner/Subject Specialist

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(Signature)

Accepted By:

_____
(For office use)

EXORDIUM

In the name of Allah, the Compassionate, the Merciful.

Praise be to Allah, Lord of Creation,
The Compassionate, the Merciful,
King of Judgment-day!

You alone we worship, and to You alone we pray for help,
Guide us to the straight path

The path of those who You have favored,

Not of those who have incurred Your wrath,
Nor of those who have gone astray.

DEDICATION

I dedicate this project to my esteemed parents, whose unwavering support and constant prayers have served as the cornerstone of my strength throughout my academic endeavors. Their encouragement has been instrumental in my pursuit of excellence.

Additionally, I dedicate this work to my distinguished instructors at the Virtual University of Pakistan, who provided me with critical knowledge and established a robust foundation that has allowed me to advance as an IT professional. Their mentorship has played a crucial role in developing my expertise in this field. I also wish to express my gratitude to my peers and friends, whose ongoing motivation and collaborative spirit have greatly enhanced my personal and professional growth. Their support has significantly enriched this journey. and established the groundwork that has enabled me to progress as an IT professional.

Their guidance has been pivotal in shaping my understanding of this field. Additionally, I wish to acknowledge my peers and friends, whose continuous motivation and collaborative spirit have significantly contributed to my growth and achievement. Their support has made this undertaking even more rewarding.

ACKNOWLEDGEMENT

First and foremost, I wish to express my profound gratitude to Almighty Allah for endowing me with the health, strength, and perseverance required to successfully complete this degree and the associated final project. It is through His blessings that I have reached this important milestone in my academic career.

I would like to extend my sincere appreciation to my supervisor, ***Abdul Majid Khokhar***, whose professional guidance and mentorship have been pivotal throughout the development of the **Woodcraft** Web Application. His extensive expertise and insights were crucial in refining the system architecture and ensuring that the project met both rigorous academic and technical standards.I would also like to acknowledge the faculty members of the ***Department of Computer Science and Information Technology*** at the ***Virtual University of Pakistan***.

Their continuous support and encouragement during my studies have significantly contributed to my academic development. Each faculty member has played a vital role in fostering an environment conducive to learning.

Lastly, I am deeply thankful to my family for their unwavering love and steadfast belief in my potential. Their encouragement and support have provided me with the motivation to overcome challenges and strive for excellence. Without their sacrifices and commitment, this achievement would not have been possible. Thank you all for your essential roles in this journey.

PREFACE

As an accomplished 8th-semester BSIT student, I am proud to present the *Woodcraft Web Application*, a project that exemplifies my technical expertise and the practical application of cutting-edge web technologies. This platform is designed to transform traditional woodwork workshops, effectively replacing outdated manual record-keeping with a streamlined, centralized digital solution that caters to the diverse needs of its users.

The **Woodcraft** skillfully addresses essential operational challenges by incorporating automated administrative approval workflows, which enhance efficiency and accelerate decision-making processes. With real-time inventory tracking, this system ensures precise monitoring of materials, enabling workshops to operate at peak efficiency.

This report details the comprehensive development of this full-stack solution, utilizing React 18 to create a sleek, high-performance **frontend** that elevates the user experience. The robust **backend** is powered by Node.js and MySQL, providing a scalable infrastructure that adapts to growing business needs. With role-based authentication integrated into the system, we ensure secure access for all users: *Admins, Customers, and Employees* alike. The addition of a "development data fallback" feature further enhances reliability, providing users with a seamless and straightforward experience as they navigate the capabilities of the Woodcraft Management System.

I am confident that this innovative solution will redefine how woodwork workshops manage their operations, setting a new standard for efficiency and effectiveness in the industry.

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CHAPTER 1

Gathering & Analyzing Info

CH # 1 :
GATHERING & ANALYZING INFO

1.1 Introduction

The **Woodcraft** is a comprehensive web-application developed to digitize and centralize the operations of a woodwork workshop. Historically, woodwork businesses have relied on manual ledger entries and fragmented communication, which leads to inefficiencies in inventory tracking and order delays.

This system leverages a modern tech stack—React 18 for the frontend and Node.js with MySQL for the backend—to provide a role-aware platform for Administrators, Customers, and Employees to interact seamlessly.

1.2 Purpose

The primary purpose of this phase was to identify the core challenges of the woodwork industry and define the functional requirements needed to solve them. The system aims to automate the transition from a manual workflow to a digital one, ensuring that:

- Administrators can approve new users and oversee the entire business lifecycle.
- Customers can place commissions and track their furniture orders in real-time.
- Employees can log work hours and view assigned tasks efficiently.

1.3 Scope:

The **Woodcraft** web application is designed to **digitize and streamline the operations of woodwork businesses** by integrating multiple operational processes into a single web-based platform. Its main objectives are:

- **Centralized Management:** Combine customer orders, inventory, employee tasks, billing, and reporting in one platform.
- **Efficiency Improvement:** Reduce manual errors and delays caused by traditional record-keeping.

The **scope** of the system includes the automation of order workflows from the moment a customer places a request until the final delivery of the product.

- ✓ It ensures that each step, such as cost estimation, assignment of workers, material usage, and payment recording, is performed accurately and transparently.
- ✓ The system will support real-time updates, enabling customers to track order progress and administrators to make informed decisions based on up-to-date information.

The system focuses on improving *productivity, reducing manual errors, and centralizing information within the woodcraft business****, without extending to external marketplace or third-party integrations.***

1.4 Definitions, Acronyms And Abbreviations

- **SRS:** Software Requirements Specification.
- **RBAC:** Role-Based Access Control.
- **UI/UX:** User Interface and User Experience.
- **CRUD:** Create, Read, Update, and Delete operations.
- **JWT:** JSON Web Token (used for secure authentication).
- **ORM:** Object-Relational Mapping.
- **Vite:** A build tool that provides a faster development experience for modern web projects.

1.5 Project Requirements

1.5.1 Functional Requirements:

Functional requirements define what the system must do. The Woodcraft system includes 5 major functional areas:

1. Customer & Order Management:

1.1 Customer Management

- The system must allow *registration of new customers*.
- Customers must be able to *log in* using secure credentials.
- Admin must be able to *view, update, edit, and search* customer profiles.
- Customer details must include name, contact, address, and order history.

1.2 Order Creation & Tracking

- Admin or customer must be able to create new woodwork orders, including:
  - ✓ Furniture orders
  - ✓ Repair orders
  - ✓ Custom design orders
- The system must auto-assign an Order ID for tracking.
- Each order must have a status workflow:
  - ✓ Pending
  - ✓ InProgress
  - ✓ Completed
  - ✓ Delivered
- Customers must be able to view their order status in real time.

1.3 Cost Estimation

- System must generate cost estimates using:
  - ✓ Material cost (based on inventory stock and price)
  - ✓ Labor cost (hourly rate + estimated hours)
- The estimate must be viewable to both Admin and Customer before payment.

2. Inventory / Materials Management

2.1 Material Tracking

- System must allow admin to add, update, or remove materials.

- Materials include:
    - ✓ Wood types
    - ✓ Nails, glue, screws
    - ✓ Paint, polish
    - ✓ Tools and consumables

2.2 Stock Monitoring

- System must auto reduce material quantity when it is assigned to an order.
- System must send a low-stock alert to admin.
- Admin must be able to view:
    - ✓ Current stock
    - ✓ Minimum threshold
    - ✓ Supplier details

3. Employee / Labor Management

3.1 Employee Task Assignment

- Admin can assign employees to specific orders.
- System must store:
    - ✓ Employee role
    - ✓ Skills
    - ✓ Assigned tasks
    - ✓ Workload

3.2 Work Progress Management

- Employees must be able to log in.
- Employees must be able to:
    - ✓ View assigned tasks
    - ✓ Update task progress
    - ✓ Record daily work hours

3.3 Wage Management

- System must calculate wages based on recorded work hours.
- Employees must be able to view their wage summary.

4. Billing & Payment Management

4.1 Payment Handling

- System must allow admin to record:
  - ✓ Full payment
  - ✓ Partial payment
- System must connect payment with specific order IDs.

4.2 Invoice Generation

- System must auto-generate invoices after payment:
  - ✓ Customer details
  - ✓ Material usage
  - ✓ Labor charges
  - ✓ Taxes (if any)
  - ✓ Total balance due

4.3 Payment History

- Admin must be able to view all payment records.
- Customers must be able to view their payment history and download invoices.

5. Reports & Analytics

5.1 Financial Reports

- Monthly and yearly:
  - ✓ Income
  - ✓ Expenses
  - ✓ Profit/Loss

5.2 Inventory Reports

- Most used materials
- Materials running low
- Cost of materials used per order

5.3 Operational Reports

- Time taken per order
- Employee productivity report
- Order workload summary
- Customer order trends

1.5.2 Non-Functional Requirements:

Non-functional requirements define how the system must perform, not what it does.

1. Performance Requirements:

- System must load all main pages within 3 seconds.
- Must handle at least 100 concurrent users
- Database queries must execute efficiently.

2. Security Requirements:

- Use encrypted passwords (BCrypt/SHA).
- Role-based access control (Admin, Customer, Employee).
- Sensitive data must be accessible only to authorized users.

3. Reliability Requirements:

- System uptime should be 99%.
- Automatic daily backups required.
- System must recover after failures without data loss.

4. Usability Requirement:

- Interface must be simple, intuitive, and user-friendly.
- Must be responsive (mobile/tablet/desktop).
- Standard iconography and color schemes should be used across the system.

5. Scalability Requirements:

- System must support adding unlimited customers, orders, and materials.
- Backend should support horizontal scaling.

6. Maintainability Requirements:

- Code must follow modular architecture.
- Easy to update or extend system modules.
- Clear documentation for future developers.

7. Availability Requirement:

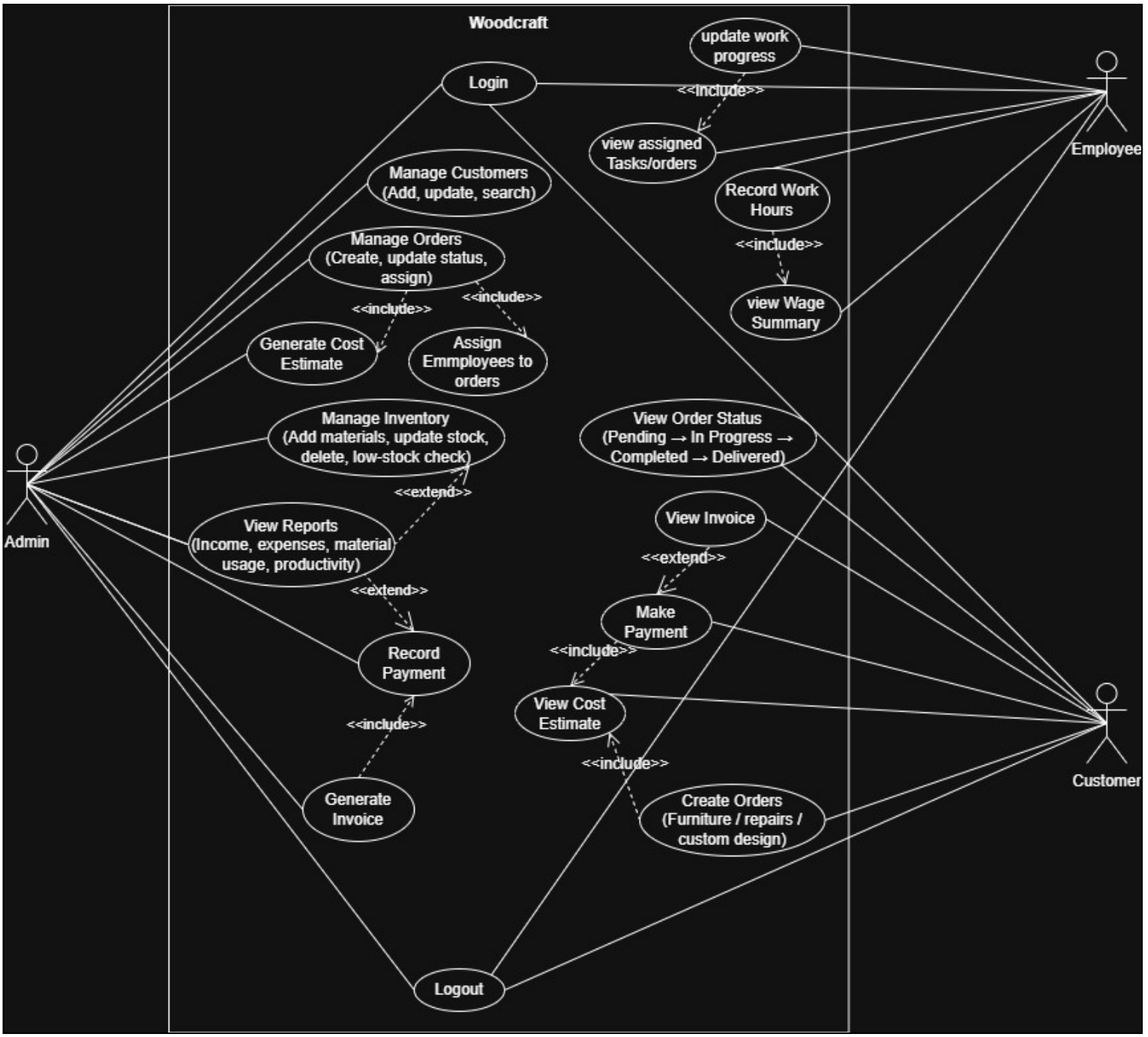
- The system must be available 24/7 with minimal downtime.
- Login and order management modules should have highest availability priority.

8. Data Integrity Requirements

- No duplicate entries for customers or inventory items.
- Stock updates must be accurate and synchronized.

1.6 Use Cases And Usage Scenarios

1.6.1 Use Case Diagrams:



1.6.2 Usage Scenarios:

1. Login/Register:

Field	Details
Use Case ID	1
Use Case Title	**Login/Register**
Actor(s)	Customer, Employee, Admin
Description	User enters valid credentials to access the system.
Pre-Conditions	User must be registered in the system. Internet available.
Post-Conditions	User successfully logged in and redirected to dashboard.
Main Actions	1. User enters username/password. 2. System validates credentials. 3. System grants access.
Alternative Paths	User selects “Forgot Password”.
Exceptions	• Invalid password. • Account locked.
Author	BC220401664

2. Logout

Field	Details
Use Case ID	**2**
Use Case Title	**Logout**
Actor(s)	Customer, Employee, Admin
Description	User exits system securely.
Pre-Conditions	User must be logged in.
Post-Conditions	Session is closed and user returns to login page.
Main Actions	1. User clicks logout. 2. System clears session. 3. User is redirected.
Alternative Paths	None
Exceptions	Session timeout before logout.
Author	BC220401664

CUSTOMER USE CASES

3. Create Order

Field	Details
Use Case ID	**3**
Use Case Title	**Create Order**
Actor(s)	Customer
Description	Customer creates a new woodwork order (furniture, repair, custom).
Pre-Conditions	Customer must be logged in.
Post-Conditions	Order is recorded and pending for review.
Main Actions	1. Customer selects “Create Order”. 2. Enters details. 3. Uploads design (optional). 4. Submits order.
Alternative Paths	Customer cancels order creation.
Exceptions	Required details missing.
Author	BC220401664

4. View Order Status

Field	Details
Use Case ID	**4**
Use Case Title	**View Order Status**
Actor(s)	Customer
Description	Customer checks progress of submitted order.
Pre-Conditions	Must have at least one order.
Post-Conditions	Status displayed successfully.
Main Actions	1. Customer opens order list. 2. System shows order progress.
Alternative Paths	No orders found.

Exceptions	Order data unavailable.
Author	BC220401664

5. View Cost Estimate

Field	Details
Use Case ID	**5**
Use Case Title	**View Cost Estimate**
Actor(s)	Customer
Description	Customer views total cost of material + labor.
Pre-Conditions	Order must exist.
Post-Conditions	Estimate displayed.
Main Actions	1. Customer selects order. 2. System shows estimate.
Alternative Paths	Customer requests revised estimate.
Exceptions	Estimation service error.
Author	BC220401664

6. Make Payment

Field	Details
Use Case ID	**6**
Use Case Title	**Make Payment**
Actor(s)	Customer
Description	Customer pays full or partial payment for order.
Pre-Conditions	Order must be approved.
Post-Conditions	Payment recorded, invoice generated (include).
Main Actions	1. Customer chooses payment mode. 2. Enters payment details. 3. Confirms transaction.
Alternative Paths	Partial Payment extends this use case

Exceptions	Payment failed.
Author	BC220401664

7. View Invoice

Field	Details
Use Case ID	**7**
Use Case Title	**View Invoice**
Actor(s)	Customer
Description	Customer views auto-generated invoice.
Pre-Conditions	Payment completed.
Post-Conditions	Invoice displayed and downloadable.
Main Actions	1. Customer selects order. 2. Clicks invoice.
Alternative Paths	None
Exceptions	Invoice not generated.
Author	BC220401664

EMPLOYEE USE CASES

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8. View Assigned Tasks

Field	Details
Use Case ID	**8**
Use Case Title	**View Assigned Tasks**
Actor(s)	Employee
Description	Employee sees orders assigned by admin.
Pre-Conditions	Employee must be assigned a task.
Post-Conditions	Tasks displayed.

Main Actions	Open dashboard → View task list.
Alternative Paths	No tasks assigned.
Exceptions	Data load failure.
Author	BC220401664

9. Update Work Progress

Field	Details
Use Case ID	**9**
Use Case Title	**Update Work Progress**
Actor(s)	Employee
Description	Employee updates task progress.
Pre-Conditions	Must have a current task.
Post-Conditions	Order progress updated.
Main Actions	Select task → Update status → Save.
Alternative Paths	None
Exceptions	Invalid update.
Author	BC220401664

10. Record Work Hours

Field	Details
Use Case ID	**10**
Use Case Title	**Record Work Hours**
Actor(s)	Employee
Description	Employee logs working hours for wage calculation.
Pre-Conditions	Employee logged in.
Post-Conditions	Hours saved.
Main Actions	Enter hours → Submit.
Alternative Paths	None

Exceptions	Invalid hour format.
Author	BC220401664

11. View Wage Summary

Field	Details
Use Case ID	**11**
Use Case Title	**View Wage Summary**
Actor(s)	Employee
Description	Employee views wages calculated from hours.
Pre-Conditions	Hours must be recorded.
Post-Conditions	Wage summary shown.
Main Actions	Open wages tab → System displays summary.
Alternative Paths	None
Exceptions	No recorded hours.
Author	BC220401664

ADMIN USE CASES

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12. Manage Customers

Field	Details
Use Case ID	**12**
Use Case Title	**Manage Customers**
Actor(s)	Admin
Description	Admin registers, updates, searches customers.
Pre-Conditions	Admin logged in.
Post-Conditions	Customer records updated.
Main Actions	Search → Add → Edit → Save.

Alternative Paths	None
Exceptions	Duplicate customer record.
Author	BC220401664

13. Manage Orders

Field	Details
Use Case ID	**13**
Use Case Title	**Manage Orders**
Actor(s)	Admin
Description	Admin views, updates, approves orders.
Pre-Conditions	Order must exist.
Post-Conditions	Order updated.
Main Actions	Open order → Modify → Save.
Alternative Paths	None
Exceptions	Order ID not found.
Author	BC220401664

14. Assign Employees

Field	Details
Use Case ID	**14**
Use Case Title	**Assign Employees**
Actor(s)	Admin
Description	Admin assigns workers to customer orders.
Pre-Conditions	Order exists, employees available.
Post-Conditions	Employee-task link saved.
Main Actions	Select order → Select employee → Assign.
Alternative Paths	None
Exceptions	No available employee.

Author	BC220401664

15. Manage Inventory

Field	Details
Use Case ID	**15**
Use Case Title	**Manage Inventory**
Actor(s)	Admin
Description	Admin updates stock of materials.
Pre-Conditions	Inventory module active.
Post-Conditions	Updated stock database.
Main Actions	Add items → Update quantity → Delete.
Alternative Paths	Auto low-stock alert (include).
Exceptions	Negative stock error.
Author	BC220401664

16. Record Payments

Field	Details
Use Case ID	**16**
Use Case Title	**Record Payments**
Actor(s)	Admin
Description	Admin enters manual payments from customers.
Pre-Conditions	Order must exist.
Post-Conditions	Payment saved.
Main Actions	Select order → Enter payment → Confirm.
Alternative Paths	Partial payment.
Exceptions	Invalid amount.
Author	BC220401664

17. Generate Reports

Field	Details
Use Case ID	**17**
Use Case Title	**Generate Reports**
Actor(s)	Admin
Description	Admin views income, expenses, stock usage, employee productivity.
Pre-Conditions	Data must exist.
Post-Conditions	Report displayed.
Main Actions	Choose report type → Generate.
Alternative Paths	None
Exceptions	Report generation failure.
Author	BC220401664

SUPPORTING USE CASES

18. Generate Invoice (Include)

Field	Details
Use Case ID	**18**
Use Case Title	**Generate Invoice**
Actor(s)	System
Description	System generates invoice after payment.
Pre-Conditions	Payment successful.
Post-Conditions	Invoice created.
Main Actions	Auto-create invoice file.
Alternative Paths	None
Exceptions	Invoice generation error.
Author	BC220401664

19. Low Stock Alert (Include)

Field	Details
Use Case ID	**19**
Use Case Title	**Low Stock Alert**
Actor(s)	System
Description	System checks inventory and alerts admin.
Pre-Conditions	Item quantity reaches threshold.
Post-Conditions	Admin notified.
Main Actions	Auto-check stock → Send alert.
Alternative Paths	None
Exceptions	Alert service down.
Author	BC220401664

1.7 Development Methodology

1.7.1 Chosen Methodology:

The **Development methodology** refers to the structured approach followed to plan, design, develop, test, and deliver the project.

VU Model Used :

The **Woodcraft** project follows the **VU Process Mode**l is a **hybrid methodology** combining:

- **Waterfall's** structured, phase-by-phase documentation
- **Spiral's** iterative development and risk analysis

It follows phases **sequentially** (like Waterfall) but each phase contains **mini-iterations** (like Spiral).

Waterfall Model:

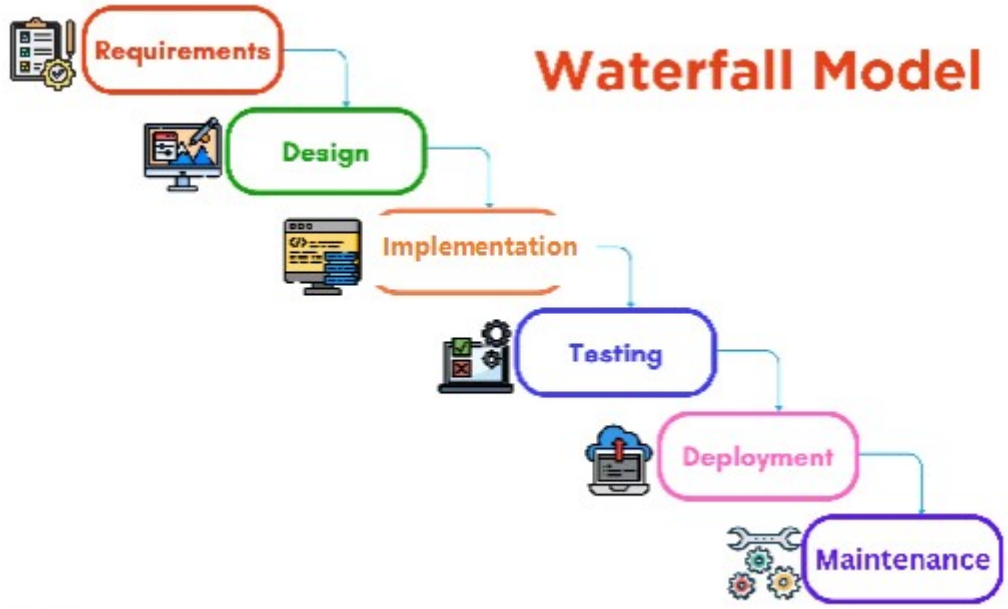
The **Waterfall Model (**part of the VU Process Model**)** is a **linear and sequential** software development process where each phase must be completed **fully** before moving to the next.

It flows downward like a waterfall through these phases:

I. **Requirements**
II. **Design**
III. **Implementation (Coding)**
IV. **Testing**
V. **Deployment**
VI. **Maintenance**

Key Point: No phase overlaps. Once a phase is finished, you move strictly forward.

Best For: Projects with clear, fixed, and well-understood requirements.



Phase of Waterfall Model:

The Waterfall Model is the oldest and most traditional software development methodology that divides the software development process into

distinct sequential ways, and each phase must be completed before moving to the next one. The sequential phases of the Waterfall Model:

1. Requirement Analysis

- Understand and identify the exact requirements your end users are expecting and document them properly
- Note down all the techniques, functions, features, and characteristics of this software
- Brainstorm, study, and analyze the demanded requirements instructed by a client
- Generate a software requirement specification (SRS), a detailed document of the software's purpose and task.

2. Design

- Transform the collected requirements into a structuralized form, which helps the developer in coding the programming language.
- Give a structured design to the whole software based on the detailed requirements collected in the first phase.
- Also, specify the software, hardware, architecture, user interface, and system requirements.

3. Implementation

- This phase is also called the coding phase
- Implement coding based on software design specification
- This phase will run unit testing to ensure each component works perfectly.

4. Testing

- All the units will be tested separately and will integrate into one system.
- The developer will run both functional and non-functional testing to ensure whether or not the system meets the requirements.

● Run a comprehensive system testing to determine the quality of the product.

● Use application lifecycle management and traceability matrix to track the testing progress.

5. Deployment

● Deploy the final product in the customer environment

● Do a sanity check of the customer environment to see it's functioning properly

6. Maintenance

● Main task is to ensure software performance in the customer environment

● Provide maintenance, operational and installation support of the software

● Note and fix issues identify by the customer

● Keep in constant update with latest update

Spiral Model:

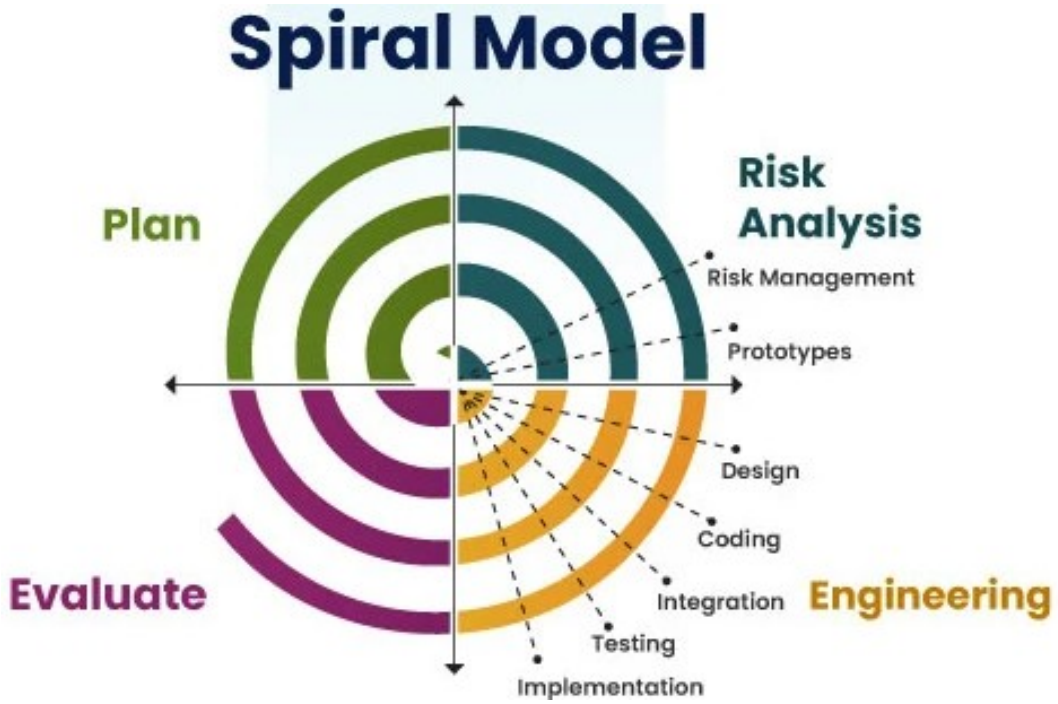
The **Spiral Model** is a **risk-driven, iterative** software development process.

The project is developed in **cycles (spirals)**, and each cycle includes:

I. **Planning**
II. **Risk Analysis**
III. **Engineering (Design/Development)**
IV. **Evaluation**

Key Point: Focuses on **continuous refinement** and **risk management** after every iteration.

Best For: Large, complex, high-risk, or unclear requirement projects.



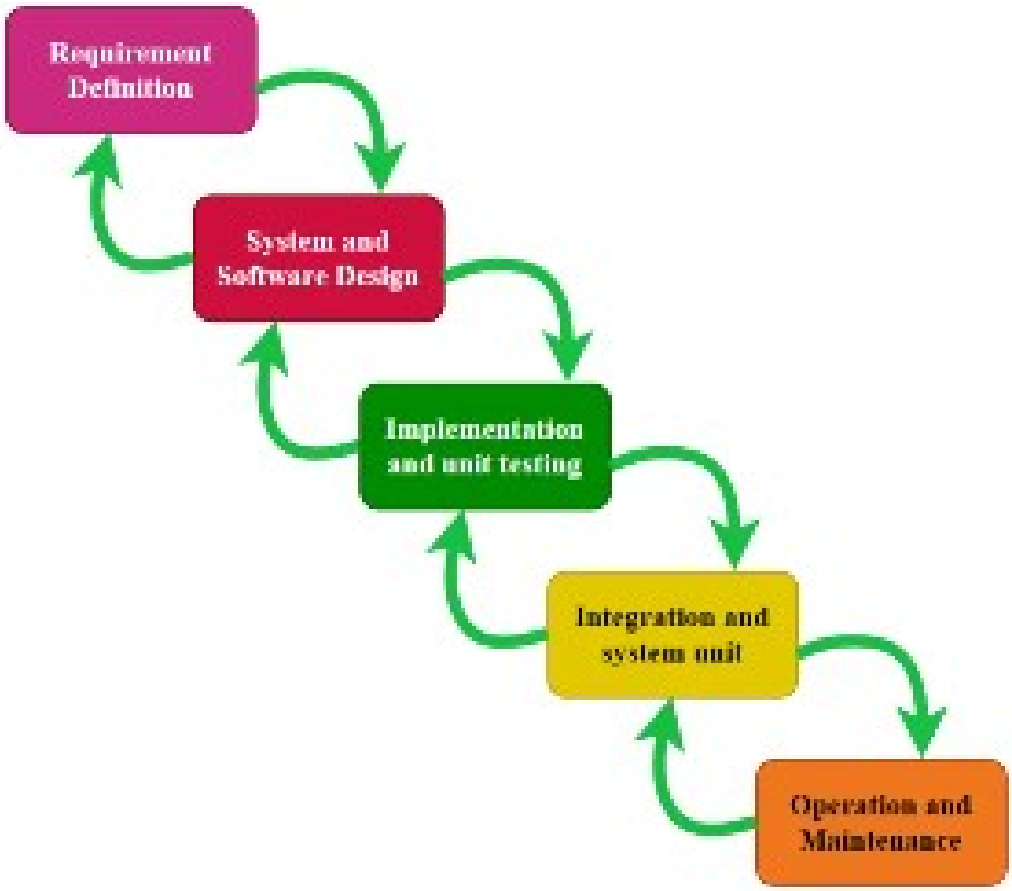
1.7.2 Reasons for Chosen Methodology:

VU Process Model:

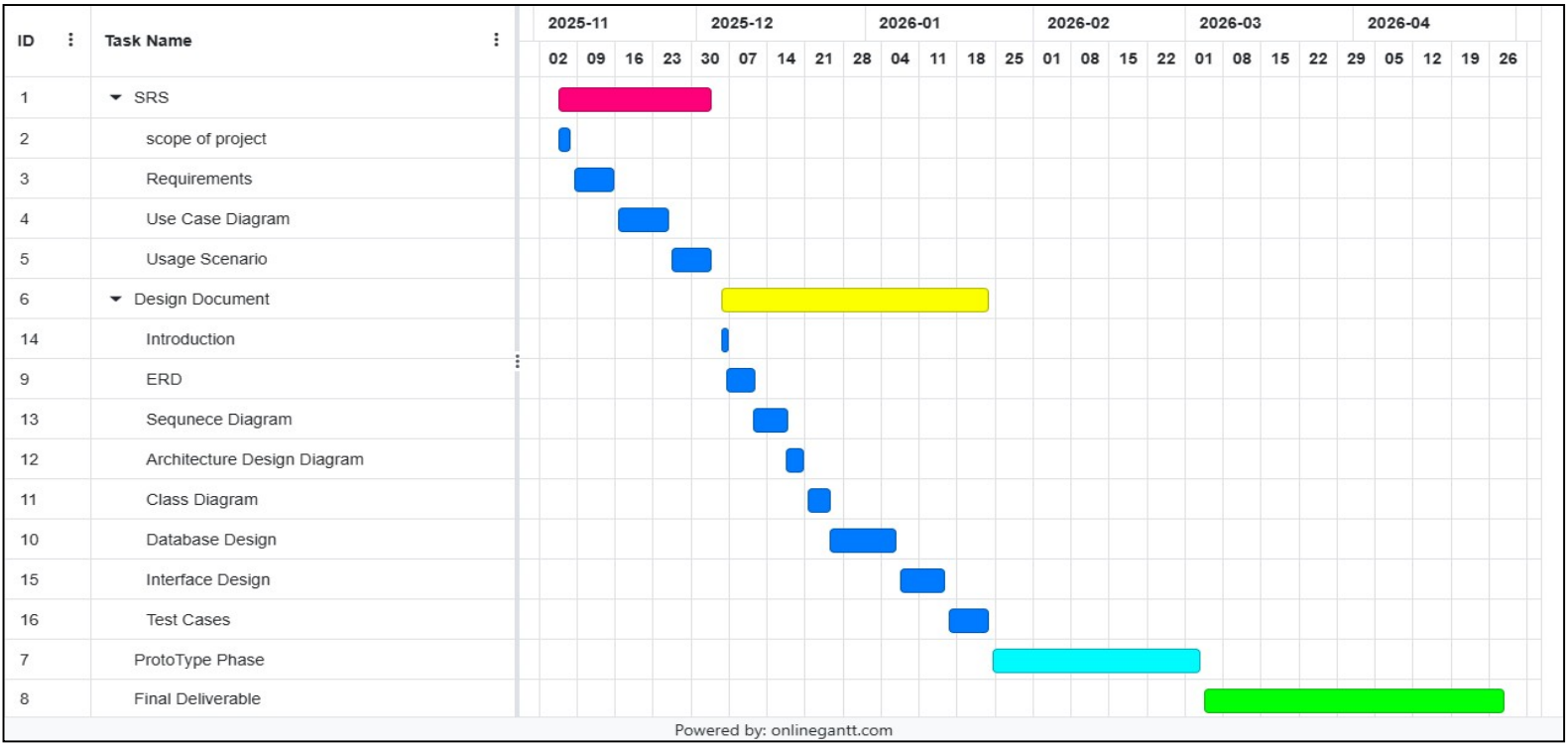
Why VU Model is Best for Woodcraft System?

Answer:

Project Requirement	Why VU Model Fits
Many interdependent modules (orders, inventory, billing, employees)	Needs structured Waterfall approach with clear phase gates.
Likely requirement changes from supervisor	Spiral allows quick modifications and re-iterations.
Complex features like cost estimation and auto stock updates	Spiral iterations reduce risk and support incremental testing.
Academic project requiring detailed documentation	Waterfall structure supports full documentation at every phase.
Need for a stable and well-tested final product	Waterfall + Spiral combination ensures reliability and quality.


Requirement Definition
System and Software Design
Implementation and unit testing
Integration and system unit
Operation and Maintenance

1.7.2 Work Plan (Gantt Chart)



1.7.2 Project Schedule (Submission Calendar)

Milestone	Description	End Date
SRS Submission	Software Requirements Specification document submitted.	04/12/2025
Design Document Submission	System design including ERD, diagrams, and DB schema submitted.	23/01/2026
Prototype	Working React frontend with mock data submitted for review.	03/03/2026
Final Deliverable	Complete Application + Final Report + Final Presentation	11/05/2026

CHAPTER 2

Designing the Project

Designing the Project

2.1 Introduction:

The design phase translates the analyzed requirements into a technical blueprint. It focuses on the structural and behavioral aspects of the Woodcraft Management System. This chapter outlines the architectural choices, such as the use of Tailwind CSS and shadcn/ui for the frontend design, and the MySQL relational schema for data persistence.

2.2 Purpose:

Purpose of this Phase in a Software Development Project:

- The Design Phase translates requirements into a practical and implementable system architecture.
- It ensures that all functional and non-functional requirements are addressed in a structured manner.
- Provides a roadmap for coding, database implementation, UI design, and testing.
- Helps in risk mitigation by identifying potential issues before development.
- Serves as a reference for stakeholders to verify that the system will meet the expected business needs.

2.3 Scope:

The Design Document is a comprehensive ***blueprint of a software system***, created after the requirements gathering and analysis phase.

- ✓ It describes how the system will be structured, how components interact, and how data flows within the system.
- ✓ For the Woodcraft web application, this document serves as a guide for developers, testers, and other stakeholders to understand

the system architecture, database structure, workflows, and functionality before implementation begins.

This document ensures that all team members have a unified understanding of the system design and reduces errors during coding and testing. It also serves as a reference for future maintenance and updates.

Design Document include the Following things:

a) Entity Relationship Diagram (ERD):

The ERD illustrates the data model of the system.

- It shows entities like Admin, Customer, Employee, Order, Material, Payment, Invoice, and Report, along with their attributes, primary keys (PK), foreign keys (FK), and relationships.

Purpose: Helps in understanding how data is stored, linked, and managed, ensuring proper database design and normalization.

---

b) Sequence Diagrams:

Sequence diagrams depict how system components interact over time to complete a particular use case.

- For Woodcraft, sequence diagrams were created for Admin processes, Employee work, Customer payments, Accounting & Reporting, and Logout.
- They include lifelines, activation bars, messages between actors and system modules.

Purpose: Clarifies workflow of operations, identifies dependencies, and ensures developers understand system behavior step-by-step.

---

c) Architecture Design Diagram

- Illustrates the high-level system structure, including layers such as Presentation (UI), Business Logic (Controller), and Data Layer (Database).
- Shows how modules communicate with each other.

Purpose: Provides a clear overview of system organization, component interaction, and scalability.

---

d) Class Diagram

- Represents ***object-oriented design*** of the system, showing classes, attributes, methods, access modifiers (public/private), and relationships like inheritance, association, and aggregation.
- For Woodcraft, ***classes include*** User (Admin/Employee/Customer), Order, Task, Inventory, Material, Payment, Invoice, Report, with proper relationships and multiplicities.

Purpose: Provides developers a blueprint for coding, ensures proper object modeling, and promotes re-usability and maintainability.

---

e) Database Design

- Includes table structures, primary keys, foreign keys, junction tables, and relationships to support the ERD.
- For Woodcraft, tables like Admin, Customer, Employee, Order, Material, Payment, Invoice, Report, and Order_Materials were designed with appropriate data types.

Purpose: Ensures data integrity, normalization, and efficient data storage, helping developers implement the database accurately.

---

f) Interface Design

- Shows UI layout, components, and interactions between users and the system.

- Covers forms, dashboards, menus, and input/output interfaces for Admin, Employee, and Customer.

Purpose: Improves usability and user experience, ensures consistent design and easier implementation for front-end developers.

---

g) Test Cases

- Includes structured test cases for each use case scenario.
- Fields include Test Case ID, Module, Objective, Preconditions, Input Data, Test Steps, and Expected Result.

Purpose: Ensures system functionality is validated against requirements, facilitates systematic testing, and identifies errors before deployment.

Benefits of this Phase in a Software Development Project:

- ❖ Clear Blueprint: Provides developers, testers, and stakeholders a visual and structured understanding of the system.
- ❖ Error Reduction: Early detection of design flaws, missing relationships, or workflow inconsistencies.
- ❖ Improved Communication: Helps teams share a common understanding of architecture, workflows, and data design.
- ❖ Faster Development: Developers can start coding with clear guidelines, reducing ambiguity.
- ❖ Better Testing: Testers can create accurate test cases based on sequence diagrams, ERD, and class diagrams.
- ❖ Future Maintenance: Provides documentation for updates, debugging, and scalability.

2.4 Definitions, Acronyms And Abbreviations:

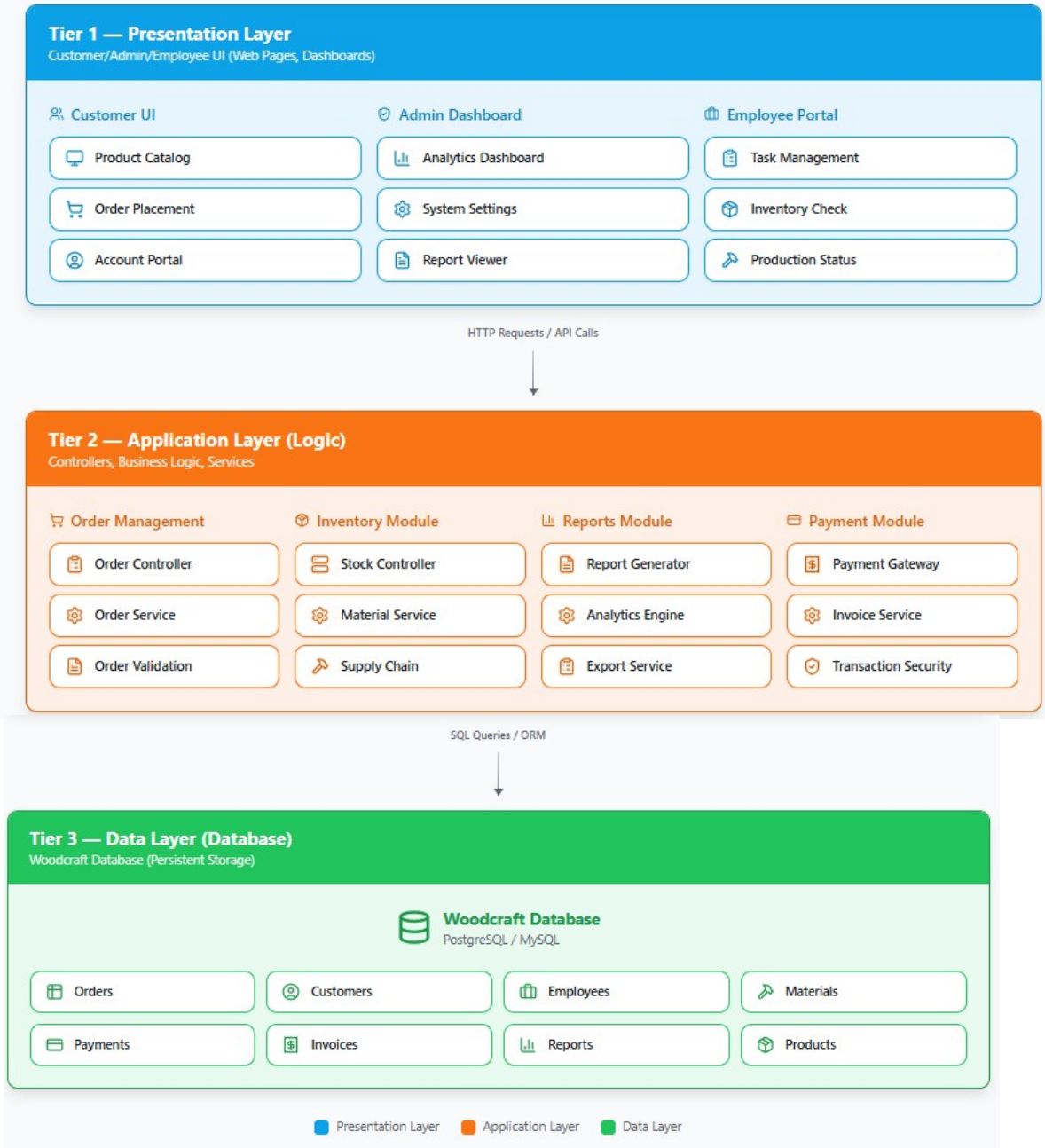
- **ERD:** Entity-Relationship Diagram.
- **API:** Application Programming Interface.

- **SQL:** Structured Query Language.
- **Bcrypt:** A password-hashing function based on the Blowfish cipher.
- **Dev Mode:** A development environment configuration using mock data.
- **Persistence:** The characteristic of state that outlives the process that created it (e.g., saving data to MySQL).
- **Backend Reference:** A implementation intended to serve as a standard or guide for the server-side logic.

2.5 Architectural Representation (Architecture Diagram)

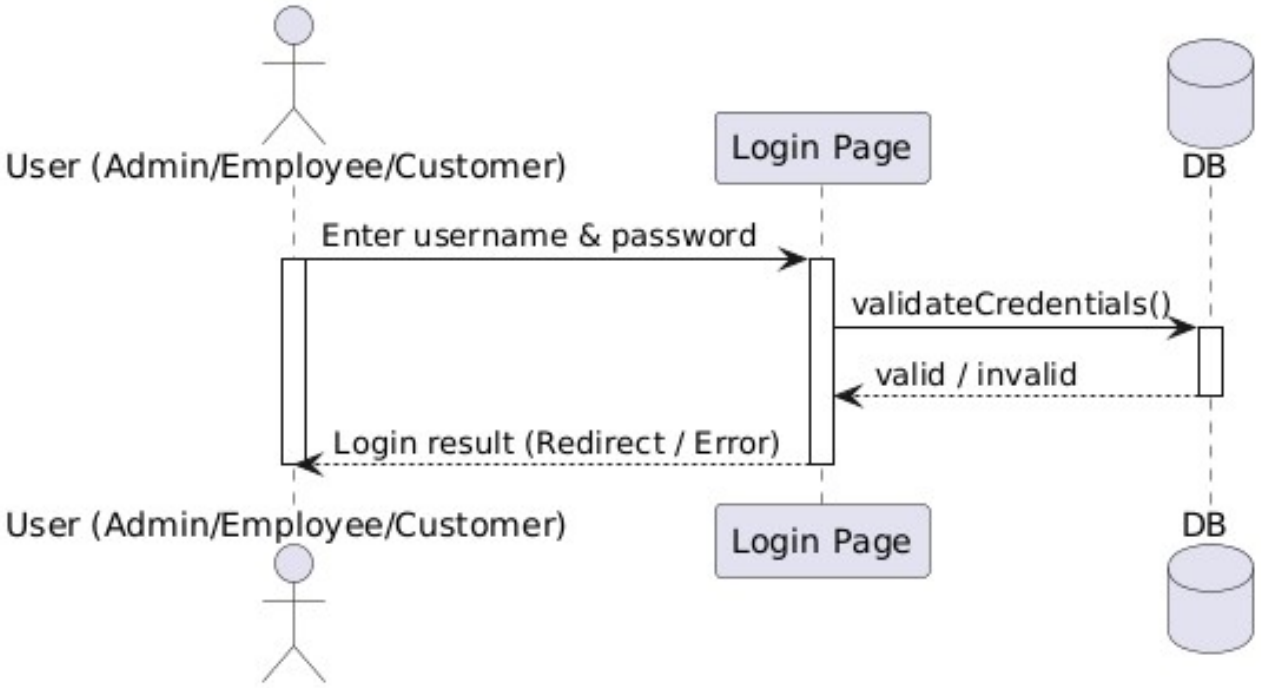
Woodcraft Management System

3-Tier Architecture

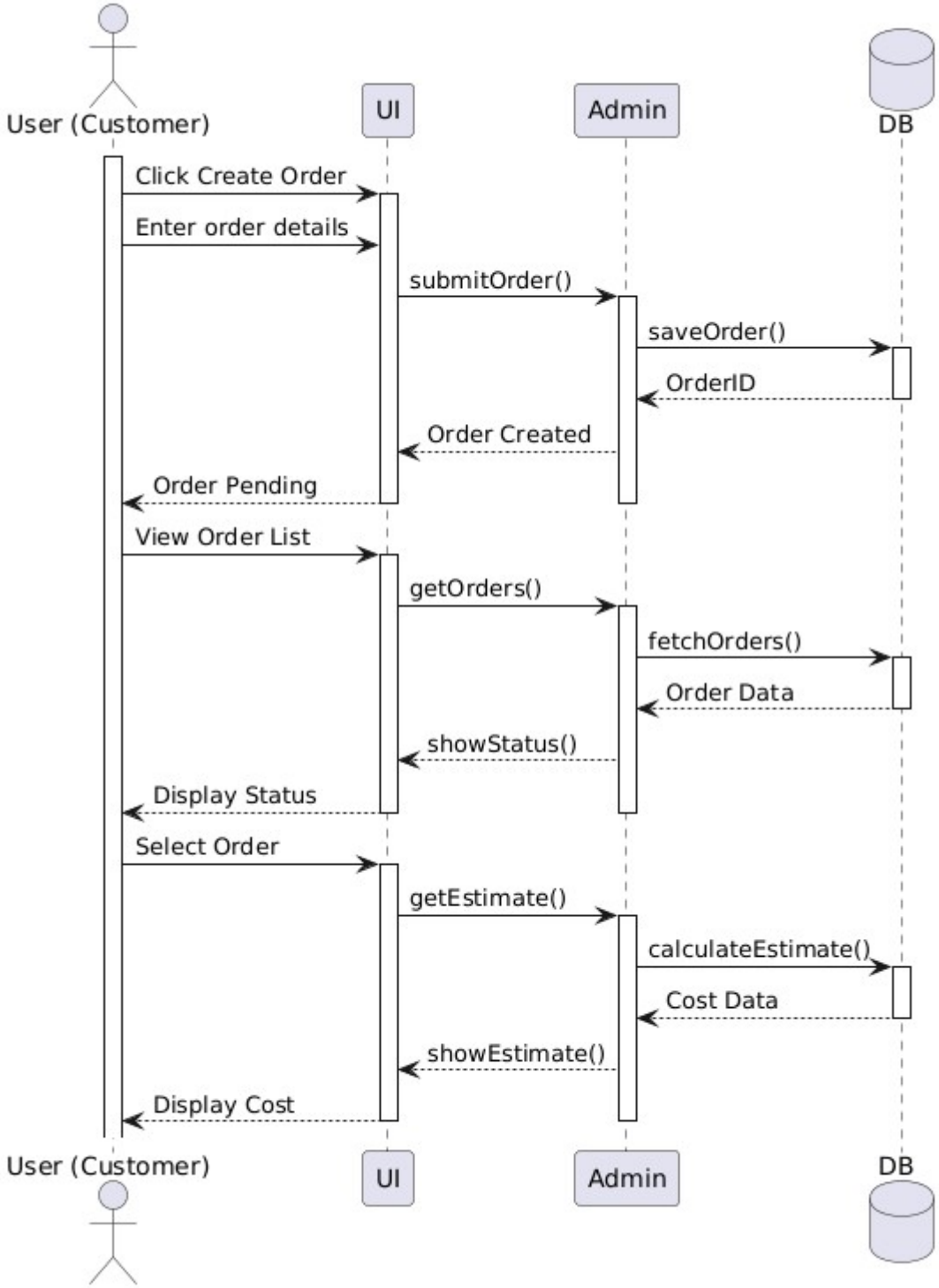


2.6 Dynamic Model: Sequence Diagrams:

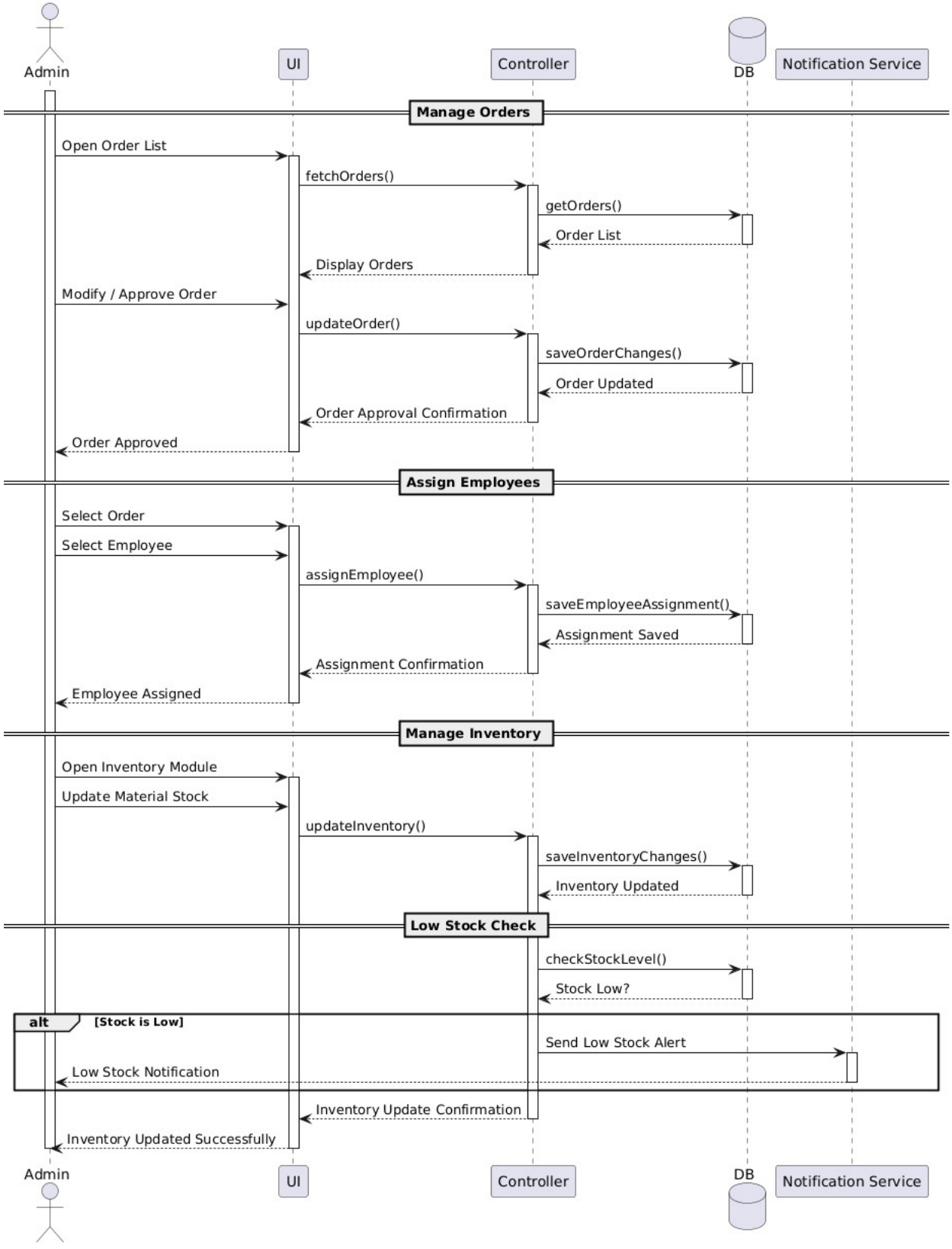
3.1 Login / Register:



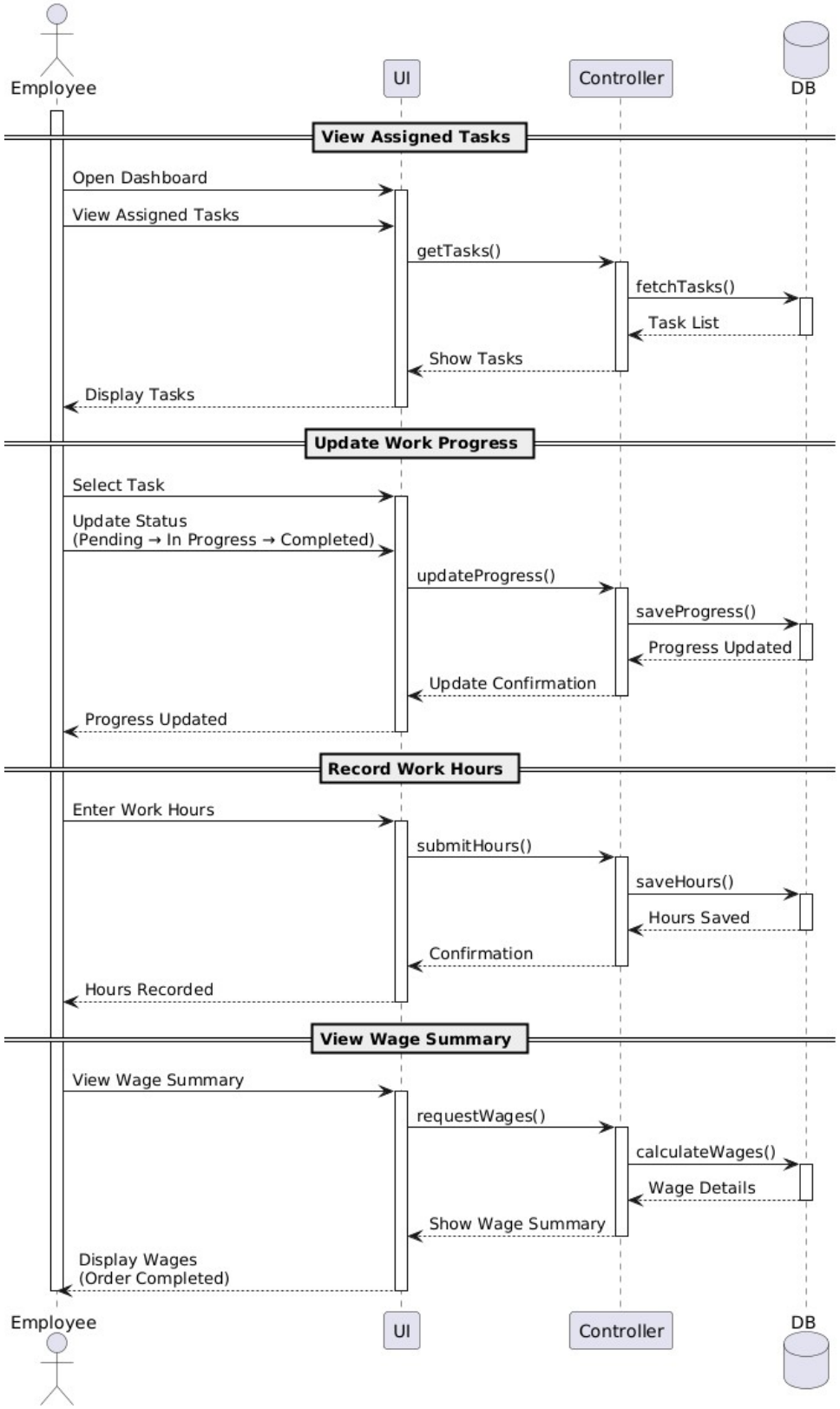
3.2 Customer initiates:



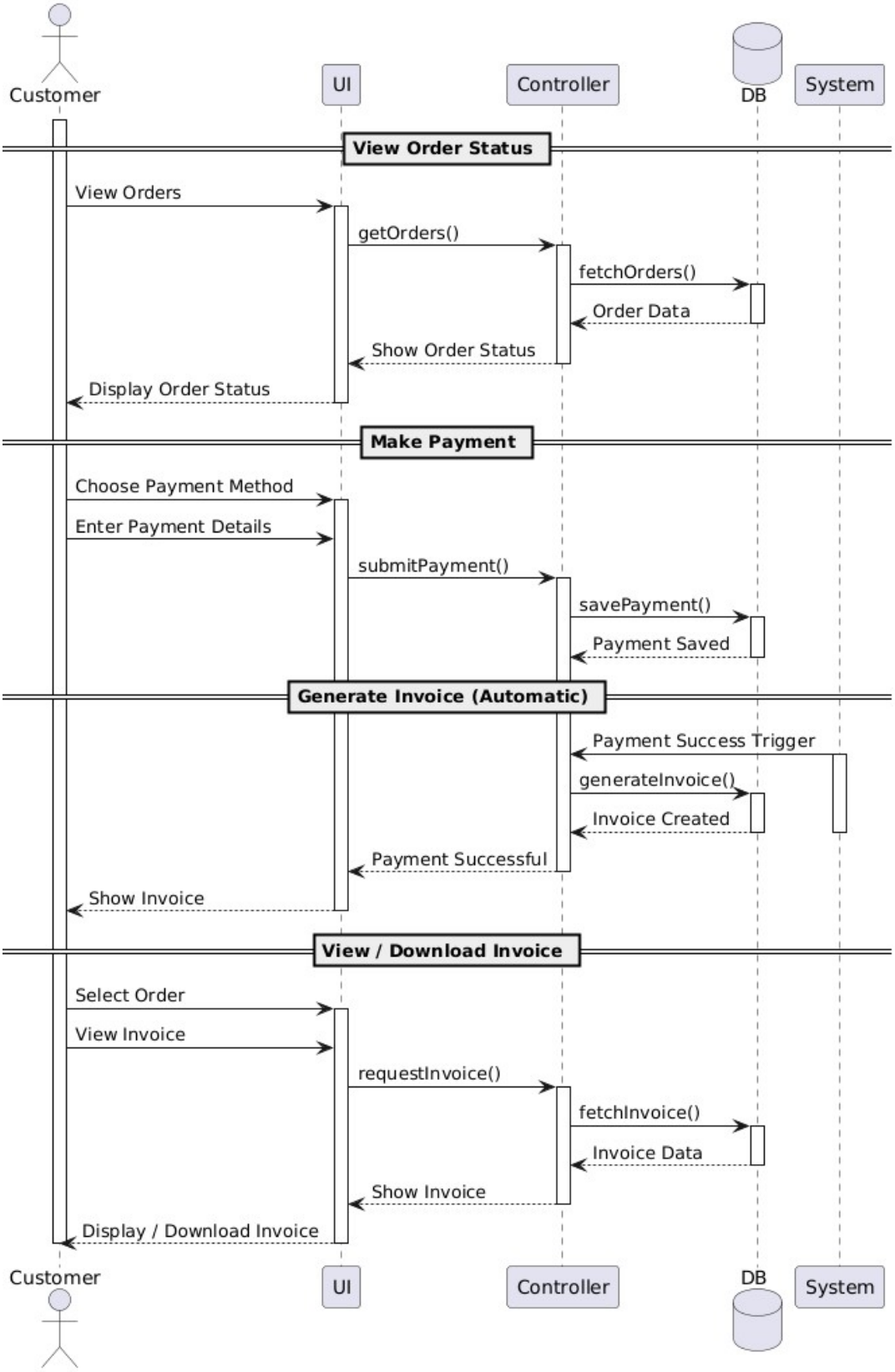
3.3 Admin Order Processing:



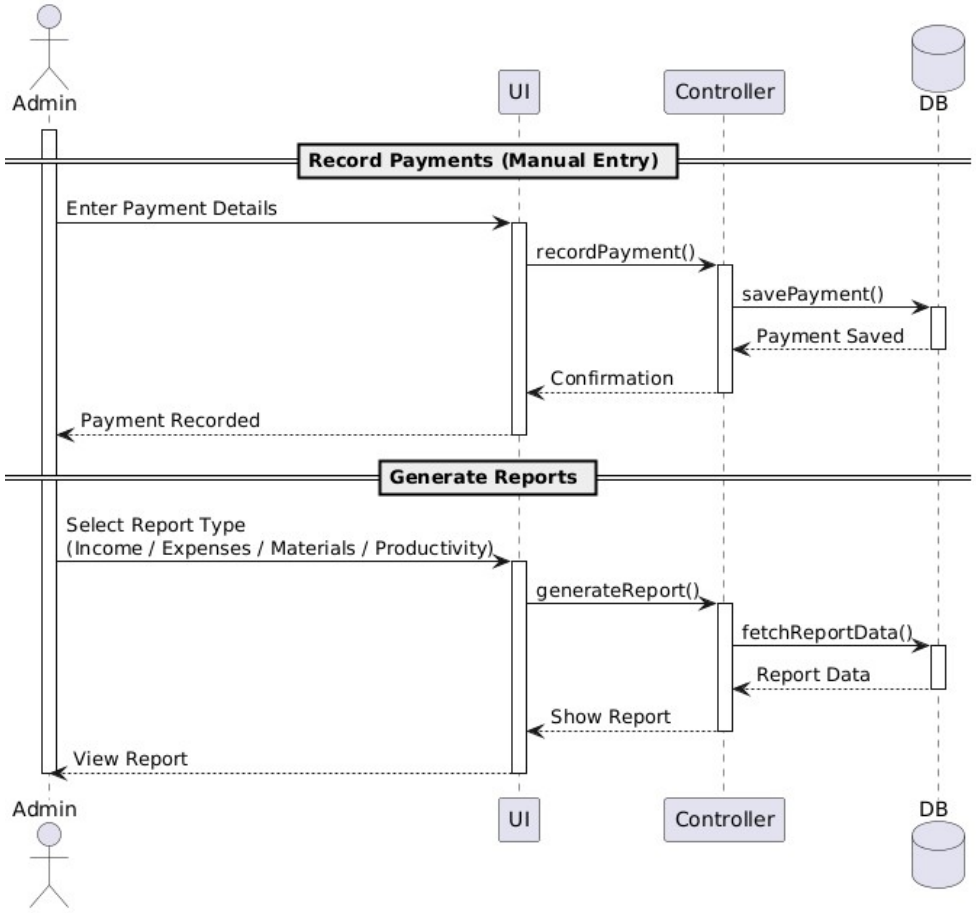
3.4 Employee performs work:



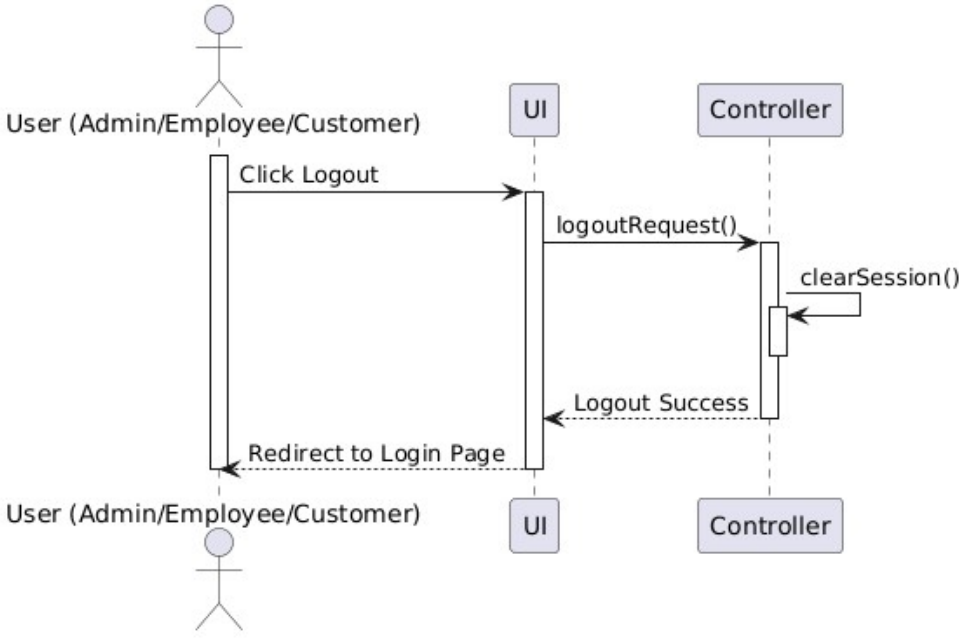
3.5 Cutomer pays:



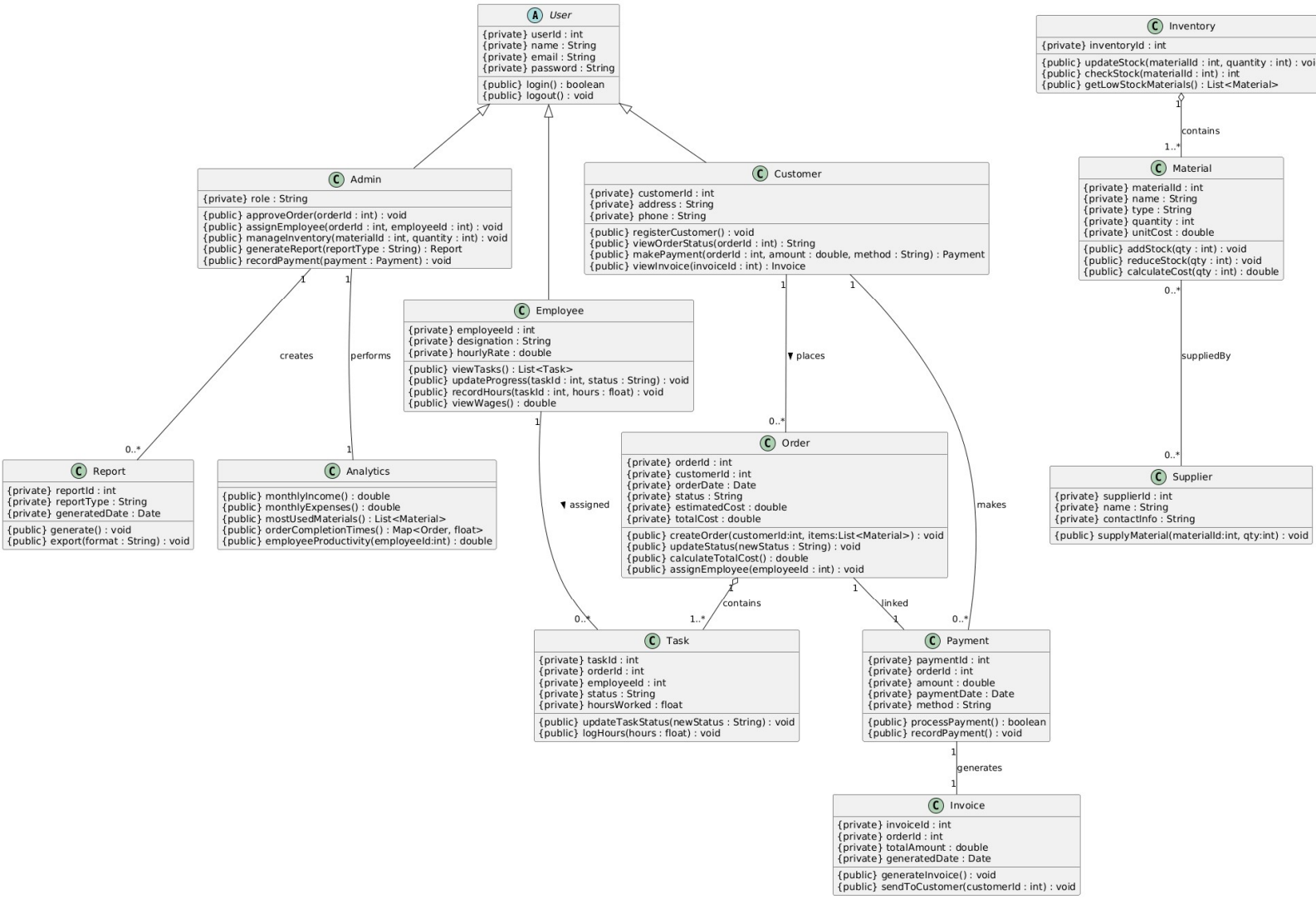
3.6 Admin accounting & reporting



3.7 Logout:



2.7 Object Model/Logical Model: Class Diagram

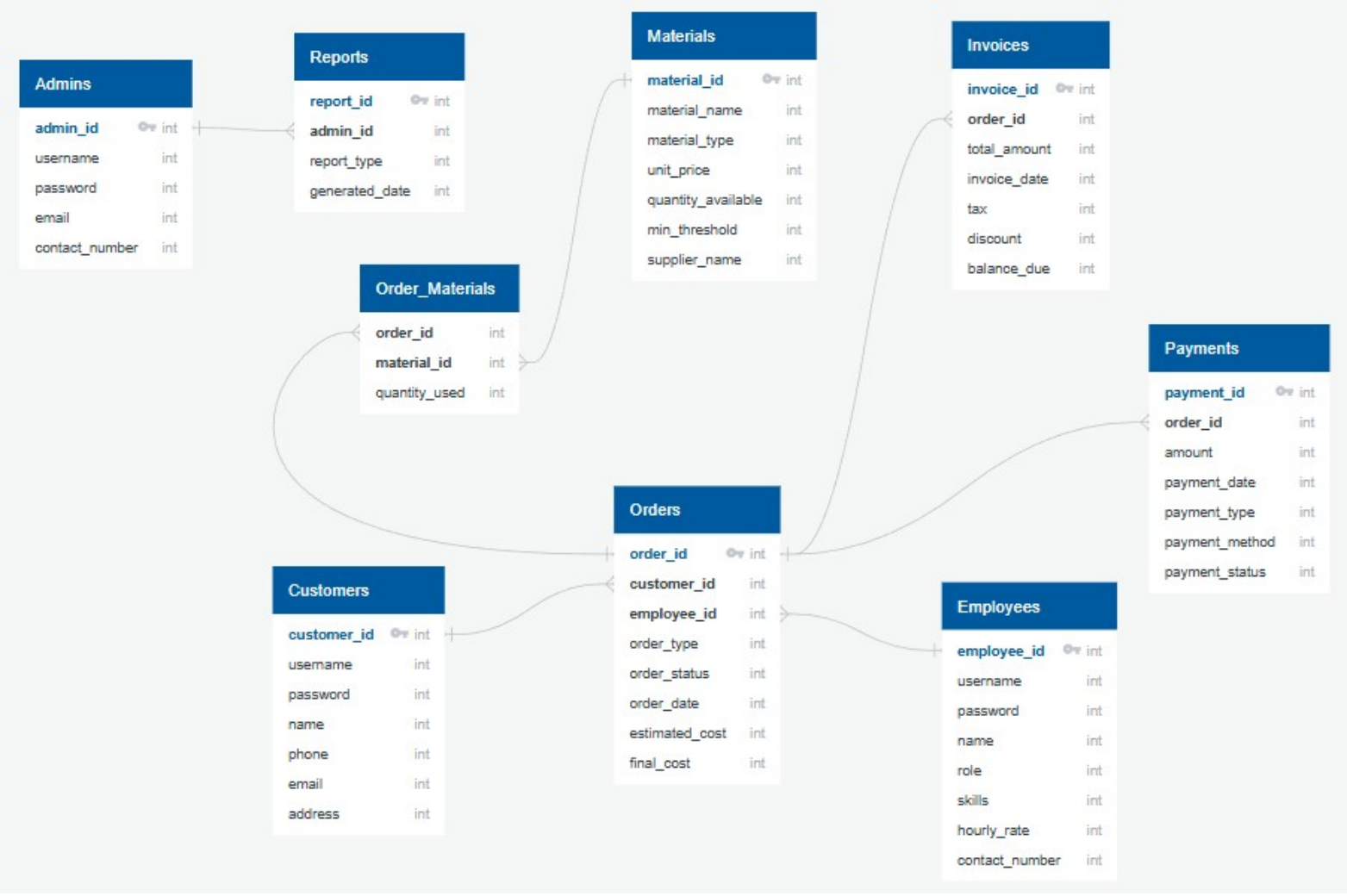


2.8 Database Model (Database Diagram)

Table Name	Field Name	Data Type	Constraints / Notes
Admin	admin_id	INT	Primary Key
	username	VARCHAR	
	password	VARCHAR	

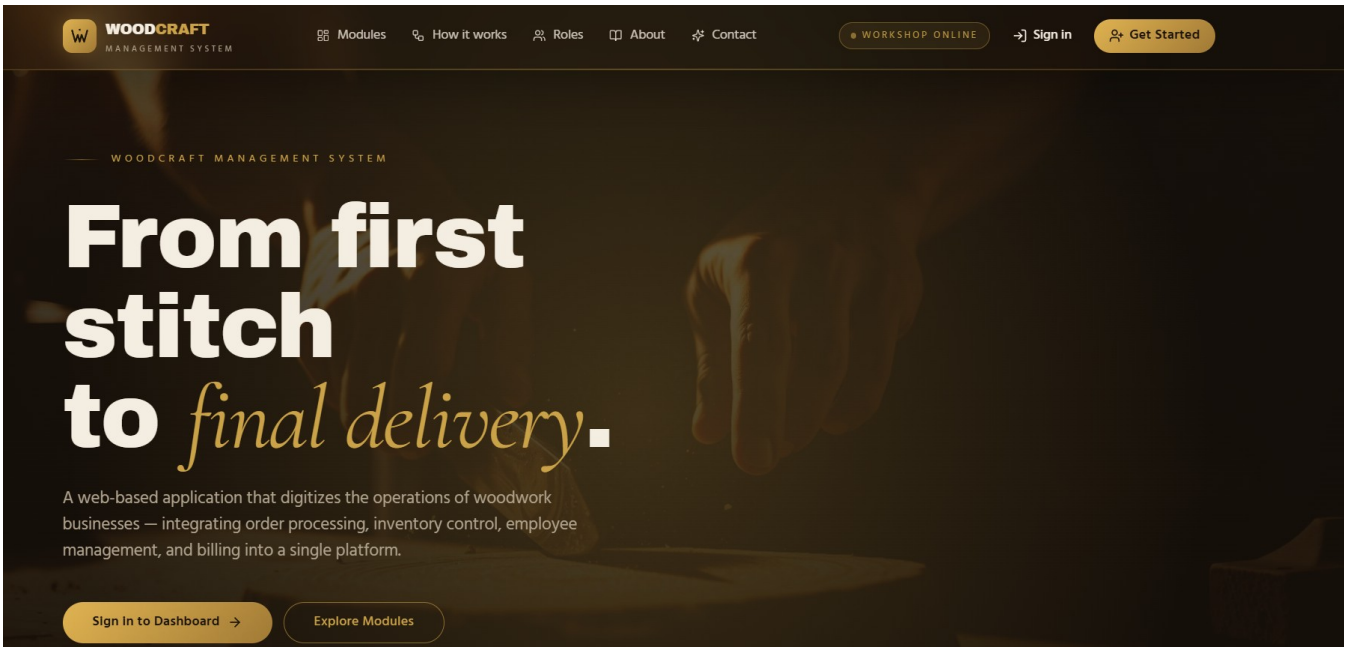
Table Name	Field Name	Data Type	Constraints / Notes
	email	VARCHAR	Unique
	contact_number	VARCHAR	
	customer_id	INT	Primary Key
	username	VARCHAR	
	password	VARCHAR	
Customer	name	VARCHAR	
	phone	VARCHAR	
	email	VARCHAR	Unique
	address	TEXT	
	employee_id	INT	Primary Key
	username	VARCHAR	
	password	VARCHAR	
	name	VARCHAR	
Employee	role	VARCHAR	
	skills	VARCHAR	
	hourly_rate	DECIMAL	
	contact_number	VARCHAR	
	material_id	INT	Primary Key
	material_name	VARCHAR	
	material_type	VARCHAR	
Material	unit_price	DECIMAL	
	quantity_available	INT	
	min_threshold	INT	
	supplier_name	VARCHAR	
	order_id	INT	Primary Key
	customer_id	INT	Foreign Key → Customer(customer_id)
	employee_id	INT	Foreign Key → Employee(employee_id)
Order	order_type	VARCHAR	
	order_status	VARCHAR	
	order_date	DATE	
	estimated_cost	DECIMAL	

Table Name	Field Name	Data Type	Constraints / Notes
	final_cost	DECIMAL	
	order_id	INT	Foreign Key → Order(order_id)
Order_Materials	material_id	INT	Foreign Key → Material(material_id)
	quantity_used	INT	
	payment_id	INT	Primary Key
	order_id	INT	Foreign Key → Order(order_id)
	amount	DECIMAL	
Payment	payment_date	DATE	
	payment_type	VARCHAR	
	payment_method	VARCHAR	
	payment_status	VARCHAR	
	invoice_id	INT	Primary Key
	order_id	INT	Foreign Key → Order(order_id)
	total_amount	DECIMAL	
Invoice	invoice_date	DATE	
	tax	DECIMAL	
	discount	DECIMAL	
	balance_due	DECIMAL	
	report_id	INT	Primary Key
Report	admin_id	INT	Foreign Key → Admin(admin_id)
	report_type	VARCHAR	
	generated_date	DATE	



2.9 Graphical User Interfaces

HomePage:



THE SYSTEM

Six modules. One *platform*.

Everything a woodwork business needs — designed to feel like one continuous tool, not six.

Customer & Order Management

Register customers and create furniture, repair, or custom-design orders. Track status from Pending to Delivered with live cost estimation.

Inventory & Materials

Manage wood, hardware, paint and polish. Stock auto-decreases on order assignment with low-stock alerts and supplier details.

Employee & Labor

Assign craftsmen to jobs, track hours and wages, and monitor performance and workload across the workshop.

Billing & Payments

Auto-generate invoices for every order, record full or partial payments, and view revenue versus expenses.

Reports & Analytics

Monthly income and expenses, most-used materials, order completion times, and employee productivity.

Role-Based Access

Separate, secure dashboards for Admin, Customer, and Employee — each user sees only what they need.

WORKFLOW

From request to *delivery*.

01

Customer requests

A customer logs in and submits a furniture, repair or custom-design commission.

02

Admin assigns

Admin allocates an artisan, picks materials and stock decreases automatically.

03

Artisan crafts

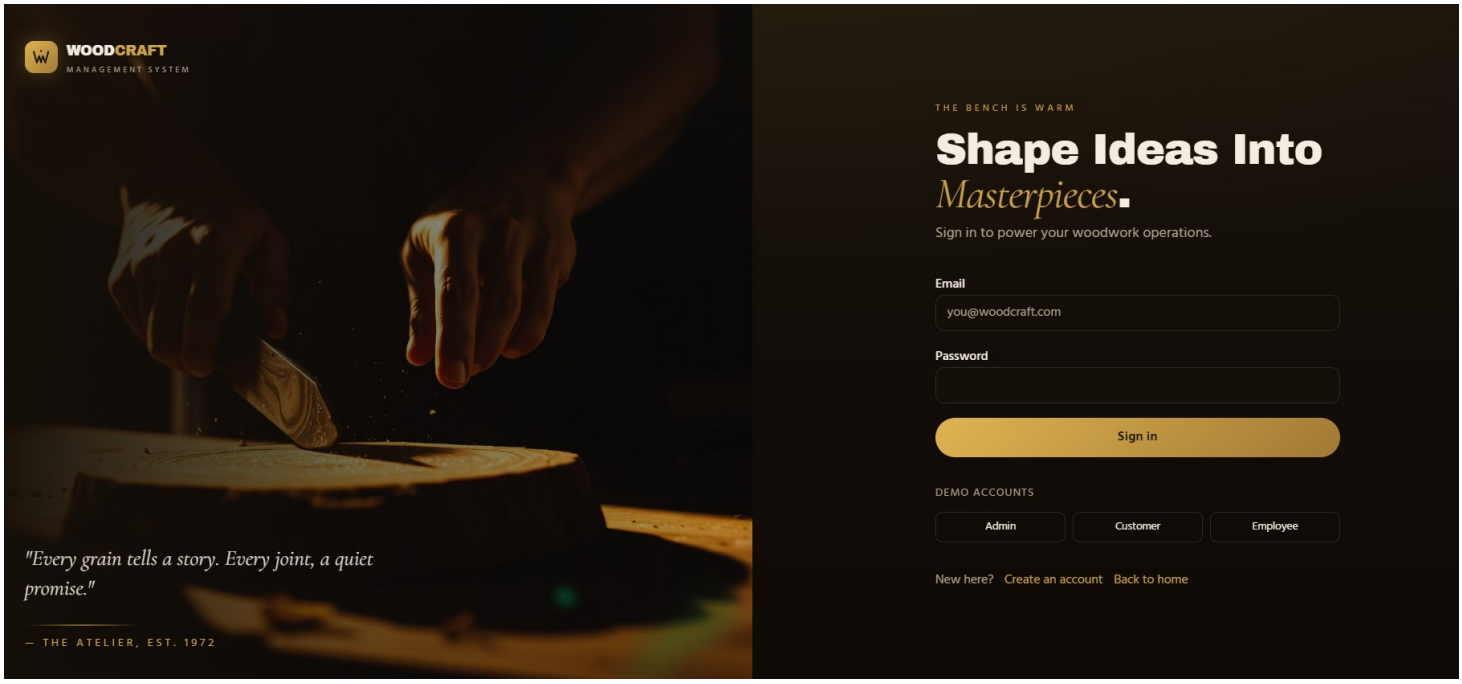
Employee logs hours and updates status as the piece moves through the workshop.

04

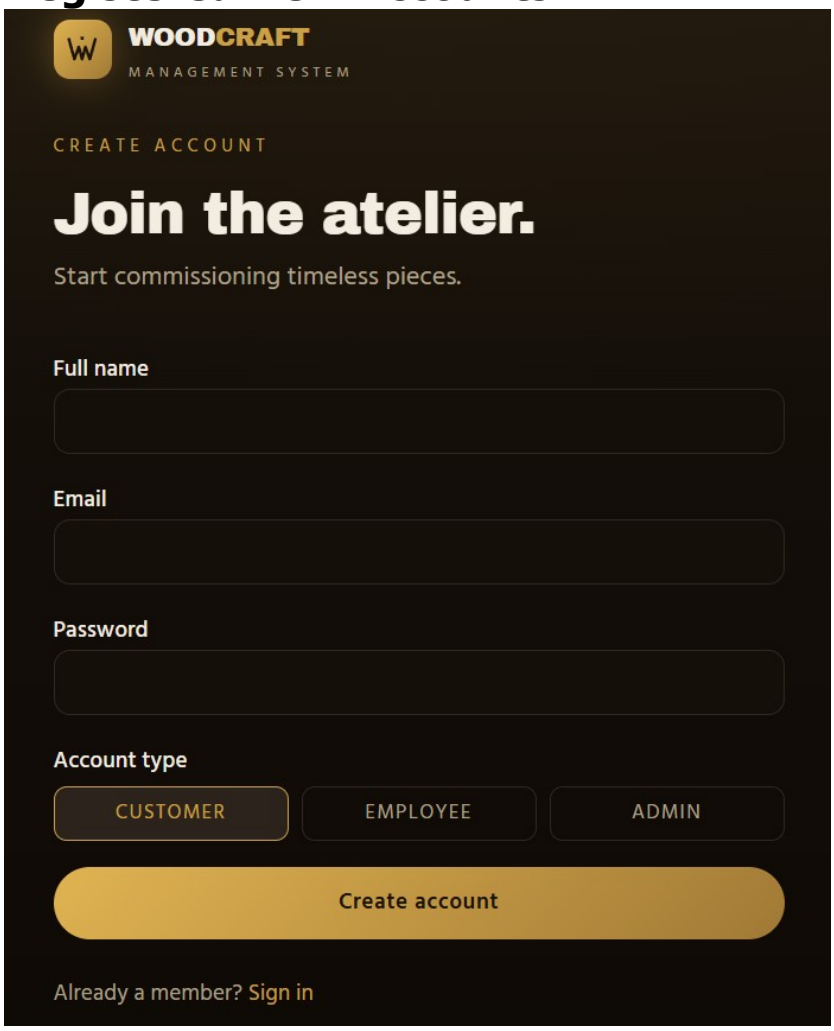
Invoice & deliver

An invoice is generated, payment is recorded and the order is marked delivered.

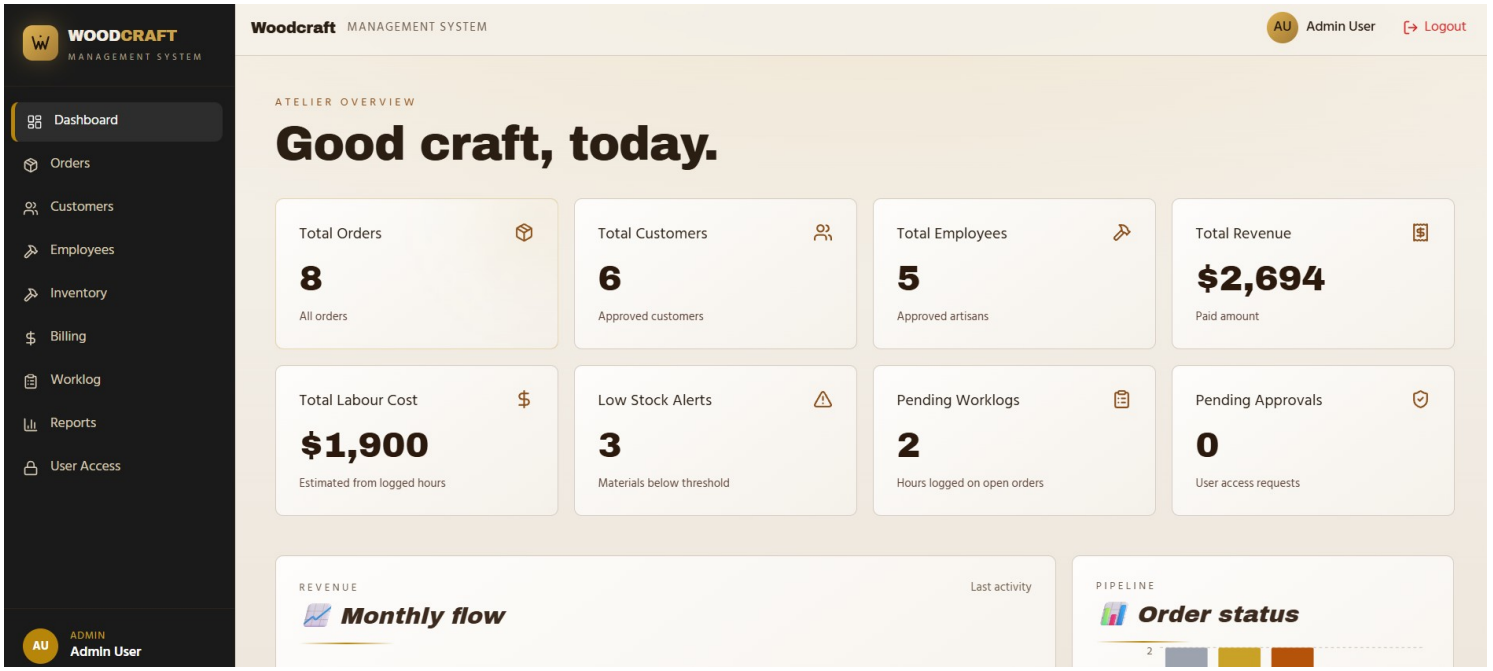
Login page:



Registered New Accounts:

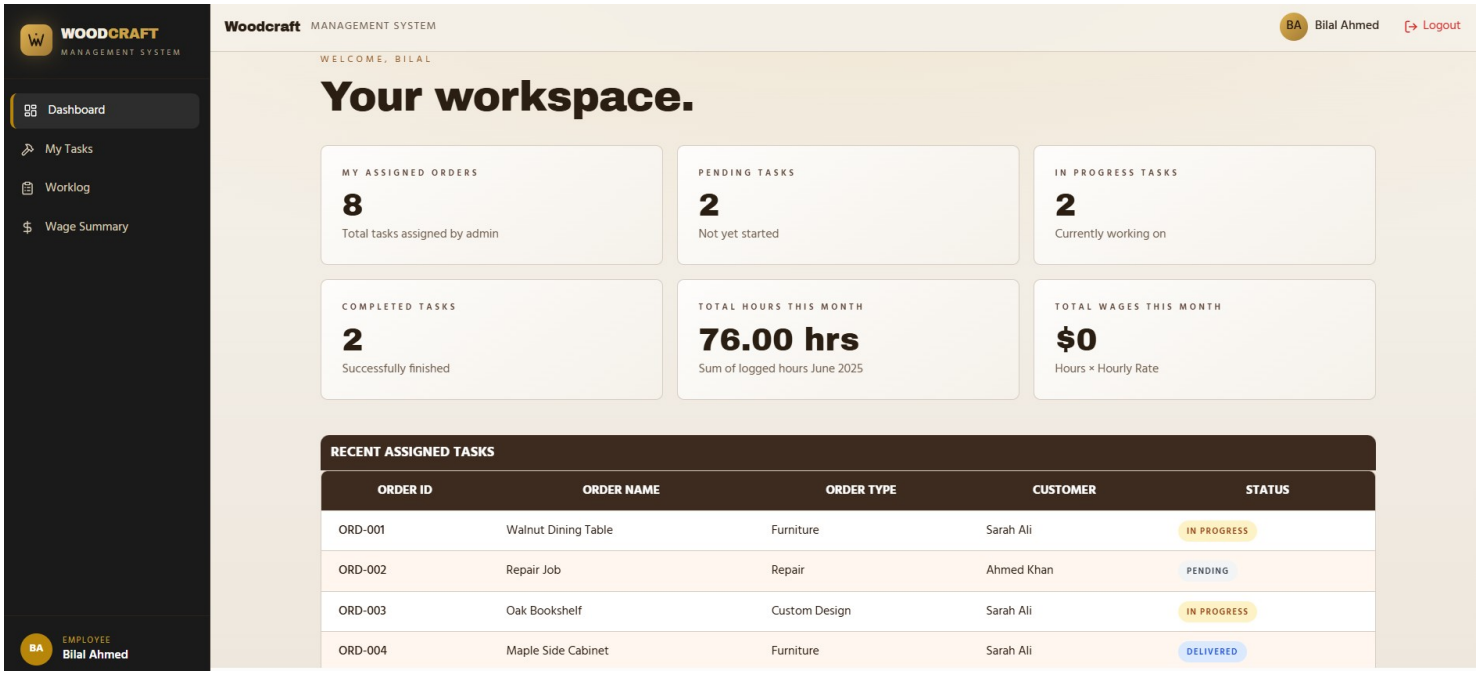


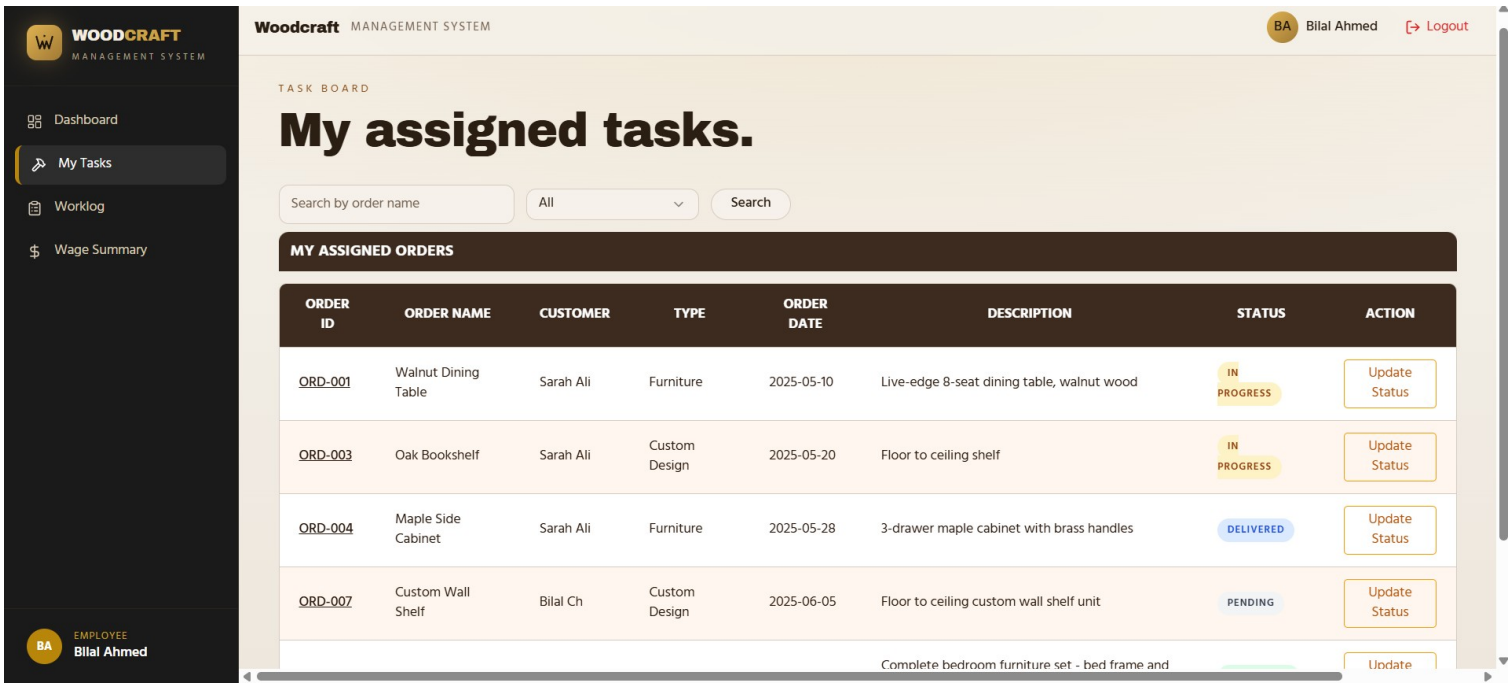
Admin login:


WOODCRAFT
MANAGEMENT SYSTEM
Dashboard
Orders
Customers
Employees
Inventory
Billing
Worklog
Reports
User Access
AU
ADMIN
Admin User
Woodcraft MANAGEMENT SYSTEM
AU Admin User
Logout
ATELIER OVERVIEW
Good craft, today.
Total Orders
8
All orders
Total Customers
6
Approved customers
Total Employees
5
Approved artisans
Total Revenue
$2,694
Paid amount
Total Labour Cost
$1,900
Estimated from logged hours
Low Stock Alerts
3
Materials below threshold
Pending Worklogs
2
Hours logged on open orders
Pending Approvals
0
User access requests
REVENUE
Monthly flow
Last activity
PIPELINE
Order status

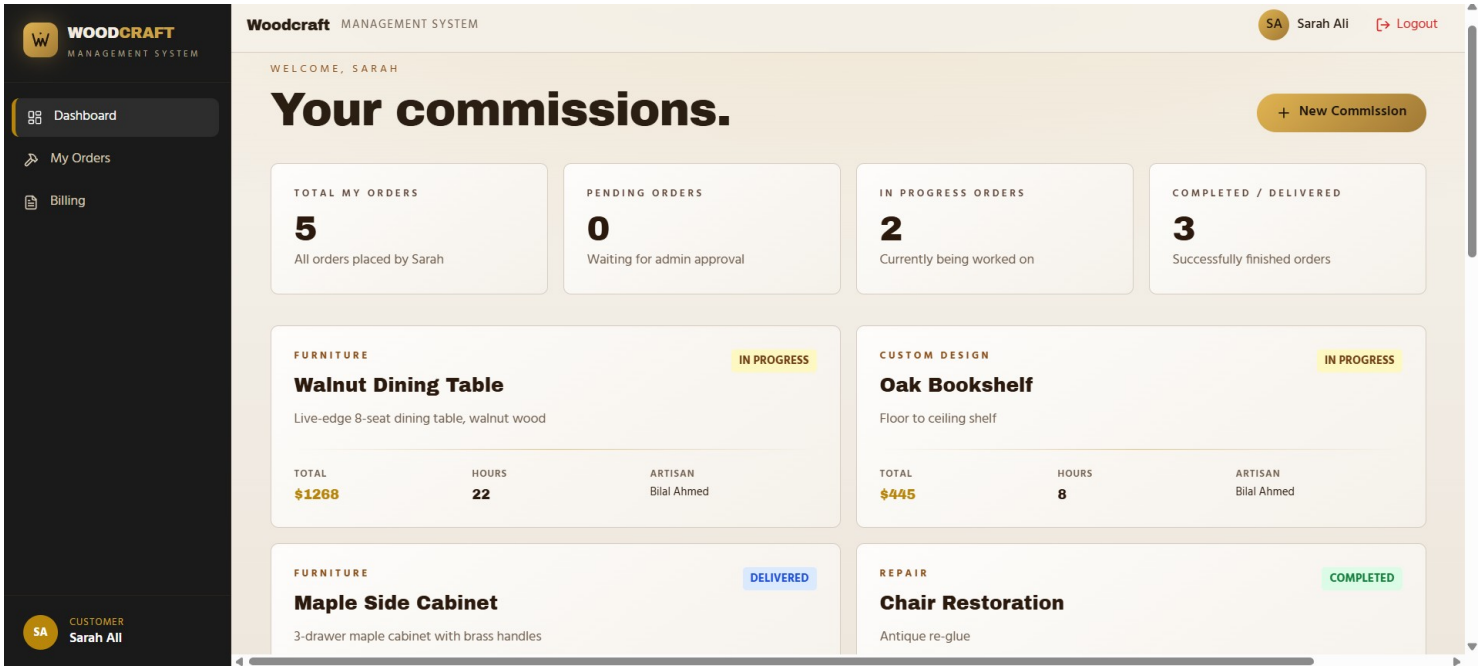

WOODCRAFT
MANAGEMENT SYSTEM
Dashboard
Orders
Customers
Employees
Inventory
Billing
Worklog
Reports
User Access
AU
ADMIN
Admin User
Woodcraft MANAGEMENT SYSTEM
AU Admin User
Logout
REVENUE
Monthly flow
Last activity
1600
1200
800
400
0
Apr
May
Jun
PIPELINE
Order status
2
1.5
1
0.5
0
Pending
In Progress
Completed
Delivered
Rejected
Low Stock Alerts
Walnut Wood
Pak Timber Co
Stock: 5
MIN: 20
Wood Varnish
Paint Plus
Stock: 2
MIN: 10
Brass Nails
Stock: 50
Recent Activity
RECENT ORDERS
Walnut Dining Table
Furniture
IN PROGRESS
Repair Job
Repair
PENDING

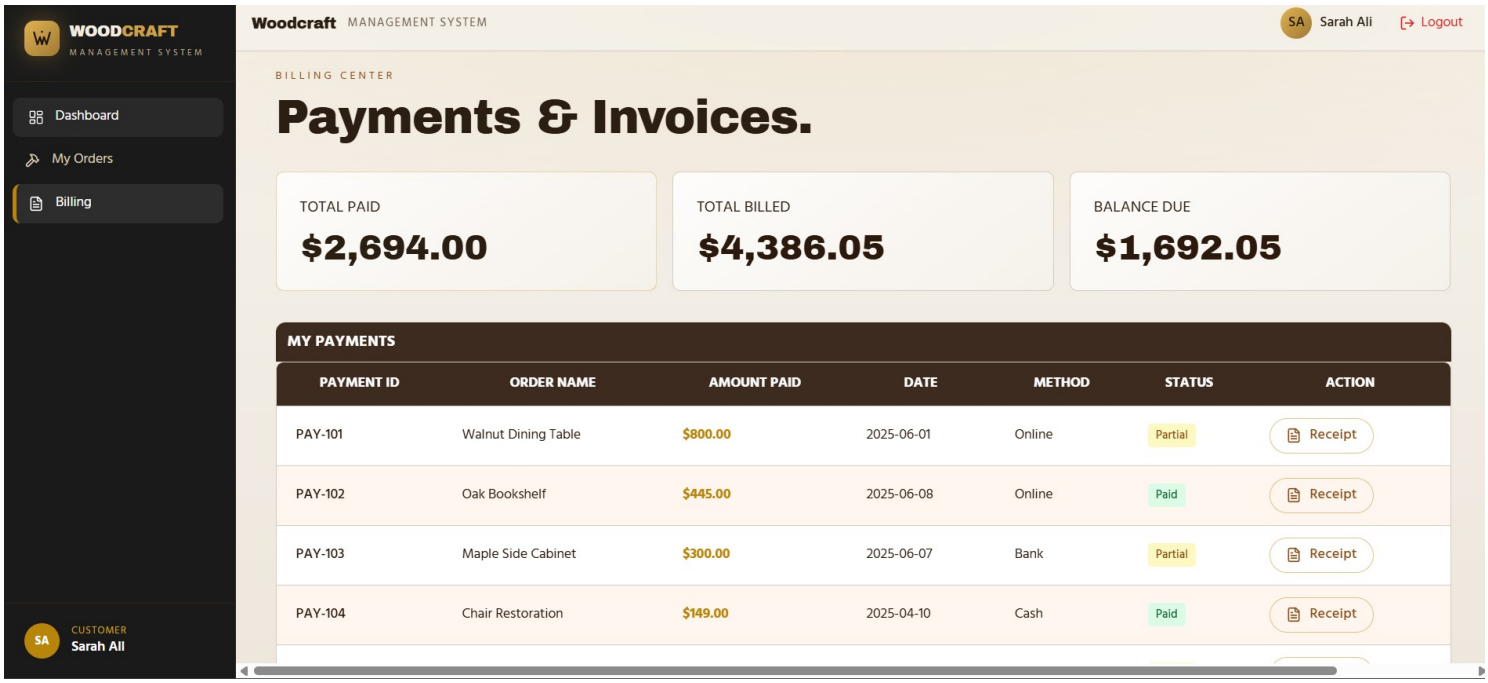
Employee login:





Customer:


WOODCRAFT
MANAGEMENT SYSTEM
Dashboard
My Orders
Billing
CUSTOMER
Sarah Ali
Woodcraft MANAGEMENT SYSTEM
SA Sarah Ali
Logout
WELCOME, SARAH
Your commissions.
+ New Commission
TOTAL MY ORDERS
5
All orders placed by Sarah
PENDING ORDERS
0
Waiting for admin approval
IN PROGRESS ORDERS
2
Currently being worked on
COMPLETED / DELIVERED
3
Successfully finished orders
FURNITURE
IN PROGRESS
Walnut Dining Table
Live-edge 8-seat dining table, walnut wood
TOTAL
$1268
HOURS
22
ARTISAN
Bilal Ahmed
CUSTOM DESIGN
IN PROGRESS
Oak Bookshelf
Floor to ceiling shelf
TOTAL
$445
HOURS
8
ARTISAN
Bilal Ahmed
FURNITURE
DELIVERED
Maple Side Cabinet
3-drawer maple cabinet with brass handles
REPAIR
COMPLETED
Chair Restoration
Antique re-glue

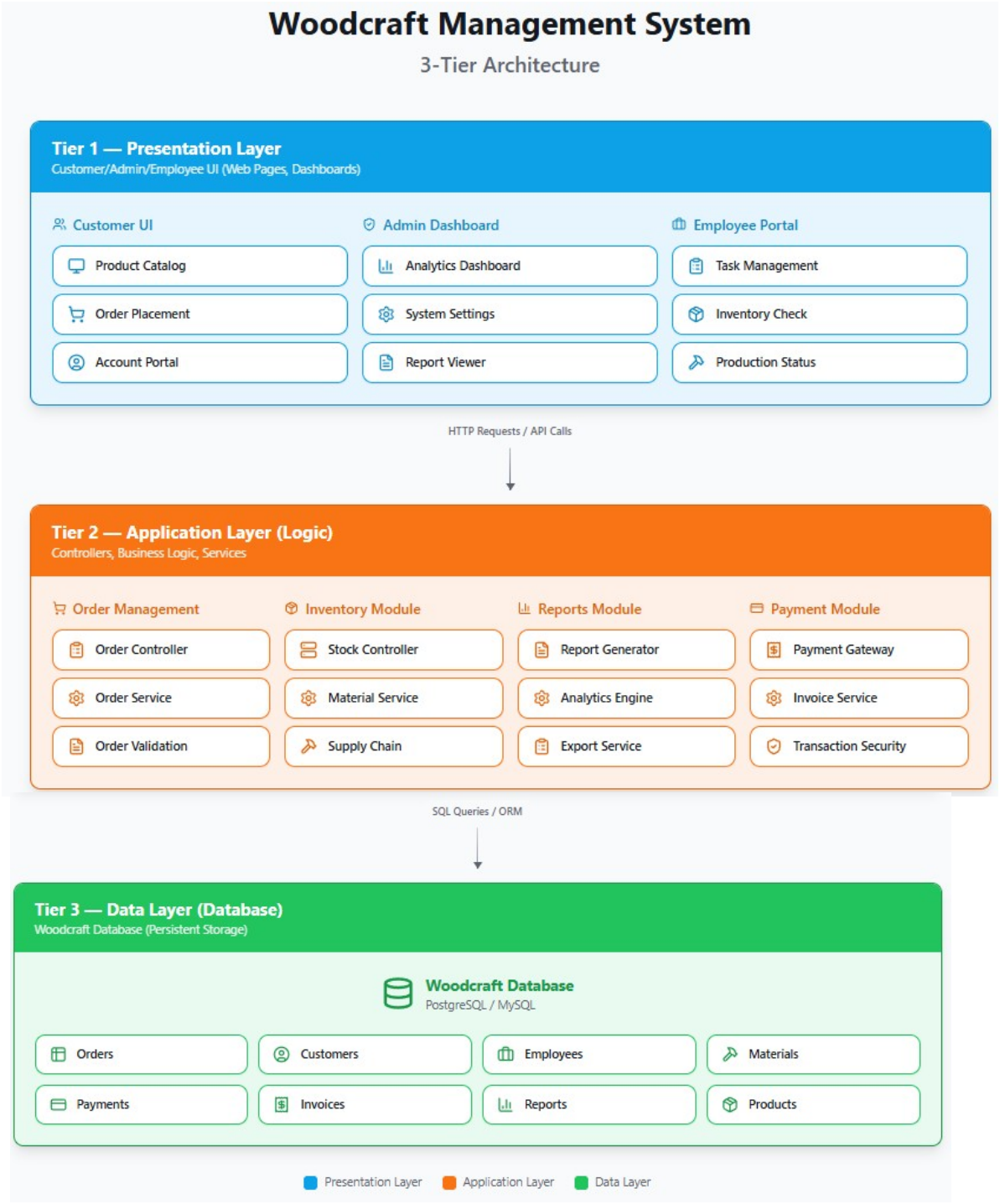

WOODCRAFT
MANAGEMENT SYSTEM
Dashboard
My Orders
Billing
CUSTOMER
Sarah Ali
Woodcraft MANAGEMENT SYSTEM
SA Sarah Ali
Logout
BILLING CENTER
Payments & Invoices.
TOTAL PAID
$2,694.00
TOTAL BILLED
$4,386.05
BALANCE DUE
$1,692.05
MY PAYMENTS
PAYMENT ID
ORDER NAME
AMOUNT PAID
DATE
METHOD
STATUS
ACTION
PAY-101
Walnut Dining Table
$800.00
2025-06-01
Online
Partial
Receipt
PAY-102
Oak Bookshelf
$445.00
2025-06-08
Online
Paid
Receipt
PAY-103
Maple Side Cabinet
$300.00
2025-06-07
Bank
Partial
Receipt
PAY-104
Chair Restoration
$149.00
2025-04-10
Cash
Paid
Receipt

CHAPTER 3

Development

CH # 3: DEVELOPMENT

3.1 DEVELOPMENT PLAN (ARCHITECTURE DIAGRAM):



REFERENCES

1. **Virtual University of Pakistan.** (2026). *Final Project (CS619/IT619) Guidelines and Documentation Templates*.
2. **React 18 Documentation.** (2026). *Modern Web Development with React and Vite*. https://react.dev/
3. **Tailwind CSS.** (2026). *Utility-First CSS Framework Documentation*. https://tailwindcss.com/
4. **MySQL Reference Manual.** (2026). *Relational Database Management and SQL Syntax*. https://dev.mysql.com/doc/
5. **Node.js & Express.** (2026). *Building RESTful APIs with Express and JWT Authentication*. https://expressjs.com/

APPENDIX

Appendix A: Technical Stack Details

- **Frontend Framework:** React 18 (TypeScript), Vite, Tailwind CSS, shadcn/ui.
- **State Management & Routing:** React Query, React Router v6.
- **Backend Environment:** Node.js, Express.js.
- **Database:** MySQL (Relational storage with SQL scripts included).
- **Security:** BCrypt (Password hashing), JSON Web Tokens (JWT Authentication).

Appendix B: System Roles & Permissions

- **Admin:** Complete control over User Access (Approvals/Blocking), Inventory, Billing, and Reports.
- **Customer:** Commission management, real-time order tracking, and invoice viewing.
- **Employee:** Task management, worklog recording, and wage tracking.

Appendix C: Mock & Persistent Data

- **Local Storage Keys:** `user`, `token`, `dev_orders`, `dev_worklogs`, `dev_pending_requests`.
- **Fallback Mode:** The system supports a "DEV mode" using local mock data to ensure the UI remains functional during backend maintenance.